

Near-Balanced Permutations at Every Size

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Abstract

For $\pi \in \mathfrak{S}_N$, let $\delta_{\pi,k}$ be the largest deviation of a length- k pattern count from $\binom{N}{k}/k!$, and let $\delta_k(N)$ be its minimum over $\mathfrak{S}_N$. Beniamini, Lavee, and Linial proved $\delta_k(N) = \Omega_k(N^{k-1})$ for fixed $k \geq 4$ and asked whether this exponent is sharp for larger k . We construct an explicit family (Π_N) , independent of k , with $\delta_{\Pi_N,k} < 14k^3 N^{k-1}/(3k!)$ for all $2 \leq k \leq N$. Thus $\delta_k(N) = \Theta_k(N^{k-1})$ for every fixed $k \geq 4$. We also determine the sharp growing-pattern range of this family: relative profile-density balance holds if and only if $k = o(N^{1/3})$. At $k \sim cN^{1/3}$ the increasing pattern has relative density $e^{-c^3/6}$. The construction completes a signed two-sheet rectangle by a mechanical partial row; the proofs isolate connected coordinate collisions after cancelling the contributions of isolated pairs. Exact rectangular profiles also identify the leading discrepancy removable by bounded-prefix rows and bounded insertions. For each fixed $k \geq 4$ and aspect $\theta > 0$, a nonconstructive realization theorem attains the standard-sheet LP distance $q_k(\theta)/[2k!(k-2)!]$. Certificates determine optimal aspects for $k = 4, \dots, 8$, including a nonsquare optimum at $k = 6$. A separate explicit family with periodic row orientations and 12 or 17 insertions attains four-pattern constant $1/192$, matching a lower bound within that completion class, not over arbitrary permutations.

Keywords. permutation patterns; balanced permutations; profile discrepancy; explicit construction; Erdős–Szekeres permutation.

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1 Introduction

For pairwise distinct real numbers $z_1, \dots, z_k$, let $\text{st}(z_1, \dots, z_k) \in \mathfrak{S}_k$ be the permutation that records their relative order. If $\pi \in \mathfrak{S}_N$ and $i_1 < \dots < i_k$, the induced pattern is $\text{st}(\pi(i_1), \dots, \pi(i_k))$. Write $\#_\tau(\pi)$ for the number of k -subsets of positions inducing $\tau \in \mathfrak{S}_k$. For example, $\text{st}(4, 1, 3) = 312$. The k -profile of π is the vector of these $k!$ counts. We study

$$\delta_{\pi,k} = \max_{\tau \in \mathfrak{S}_k} \left| \#_\tau(\pi) - \frac{1}{k!} \binom{N}{k} \right|, \quad \delta_k(N) = \min_{\pi \in \mathfrak{S}_N} \delta_{\pi,k}.$$

Thus π is k -balanced precisely when $\delta_{\pi,k} = 0$.

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A uniformly random permutation has expected profile $\binom{N}{k}/k!$, while quasirandom permutation sequences approach these densities asymptotically [2, 7]. Related frameworks include permutation limits [6] and asymptotic profile symmetry [5]. Here the objective is a finite, simultaneous ℓ_∞ approximation to the uniform profile. Joy Cooper and Dukes [4, equation (1.2)] define the fuzzy lift $P_\sigma^{\uparrow n}$ as a normalized sum of order-preserving embedding matrices. They study rational combinations of these lifts that are constant matrices. The same binomial weights appear in our insertion expansion, but those formal identities do not provide the every-size discrepancy bound proved here: we realize centred corrections by unweighted points in one permutation. Exact balance is exceptionally rigid. Beniamini, Lavee, and Linial proved that no permutation is 4-balanced and obtained the lower bound $\delta_k(N) = \Omega_k(N^{k-1})$ for every fixed $k \geq 4$ [1, Theorems 6 and 7]. Their two-sided Erdős–Szekeres construction matches this order for $k = 4$ along orders $2m^2$, but the sharp exponent for each fixed $k > 4$ was left open. An explicit all-size baseline follows from Cooper’s estimate [2, Theorem 3.1 and the proof of Proposition 3.4]: for every fixed $k \geq 2$, every $N > k$, and every $\pi \in \mathfrak{S}_N$,

$$\delta_{\pi,k} \leq C_k D(\pi) N^{k-1}, \quad D(\pi) := \max_{\substack{I, J \subseteq [N] \\ I, J \text{ intervals}}} \left| \#(\pi(I) \cap J) - \frac{|I||J|}{N} \right|,$$

Here intervals are nonwrapping; D_{cyc} denotes the same maximum over cyclic intervals in $\mathbb{Z}_N$. Splitting each cyclic interval into at most two nonwrapping intervals gives $D_{\text{cyc}} \leq 4D$. Cooper’s decoupling estimate is $O_k((D_{\text{cyc}} + 1)N^{k-1})$. Taking $I = [N - 1]$ and $J = \{\pi(N)\}$ gives $D \geq (N - 1)/N$, which absorbs the additive 1 and yields the displayed finite bound. The Sós permutation $\sigma_N^{(\alpha)}$ ranking the fractional parts $\{\alpha\}, \dots, \{N\alpha\}$, with $\alpha = (\sqrt{5} - 1)/2$, has $D(\sigma_N^{(\alpha)}) = O_\alpha(\log N)$ [3, Theorem 3.24]. Hence $\delta_{\sigma_N^{(\alpha)},k} = O_{k,\alpha}(N^{k-1} \log N)$ at every size. The issue is to remove this logarithm while retaining all patterns and all permutation sizes.

Throughout, N denotes permutation size and k denotes pattern length.

Theorem 1.1. *There is an explicit family $(\Pi_N)_{N \geq 1}$, independent of k , with the following properties.*

(i) *For all $2 \leq k \leq N$,*

$$\delta_{\Pi_N,k} < \frac{14k^3}{3k!} N^{k-1}.$$

Consequently $\delta_k(N) = \Theta_k(N^{k-1})$ for every fixed $k \geq 4$.

(ii) *For $2 \leq k = k(N) \leq N$, relative profile-density balance*

$$\max_{\tau \in \mathfrak{S}_k} \left| \frac{k! \#_\tau(\Pi_N)}{\binom{N}{k}} - 1 \right| \longrightarrow 0$$

holds if and only if $k = o(N^{1/3})$. If $k/N^{1/3} \rightarrow c \in [0, \infty)$, the increasing pattern has relative density tending to $e^{-c^3/6}$; if $k/N^{1/3} \rightarrow \infty$, its relative density tends to zero.

Theorem 1.1 closes the fixed- k exponent question of Beniamini–Lavee–Linial and removes the logarithm from Cooper’s all-size comparison. Part (ii) determines exactly when this family approximates every shrinking target $1/k!$ relatively. Its threshold is specific to Π_N , not a bound for all permutation families.

The complete two-sheet rectangle is inherited from [1]. The obstacle to an every-size construction is its boundary: filling a gap of $O(\sqrt{N})$ points can affect $\Theta_k(N^{k-1/2})$ subsets, more than the allowed discrepancy. We use a partial row whose columns form a mechanical word. Sign reversal makes isolated coordinate-tie resolutions unbiased; a coupling to independent tie-breaking leaves

only interacting collisions. Prefix balance then controls the centred profile of the large clean boundary class, rather than bounding its cardinality. One northeast point handles odd sizes. This proves part (i) in Sections 2–6.

The questions of pattern growth and removable leading error are linked by the exact rectangle formula in Section 7. A connected collision expansion cancels arbitrary collections of isolated pairs and bounds the remaining relative error by $O(k^3/N)$ on complete squares. An exact increasing-pattern count gives the matching obstruction at $k \asymp N^{1/3}$. The same formula identifies three local statistics that describe the fixed- k leading vector.

Section 8 turns that vector into a completion optimum. A bounded-prefix row contributes a term depending only on the final pattern value, and boundedly many insertions contribute sums over position–value pairs. We prove that every bounded centred correction of this additive form can be realized. The remaining discrepancy is therefore exactly the ℓ_∞ distance $q_k(\theta)$ of the core vector from the additive subspace, with normalization $2k!(k-2)!$. Exact certificates show that square aspect is optimal in this class for $k = 4, 5, 7, 8$, but not for $k = 6$: its minimizing aspects are $2 - 2\sqrt{3}/3$ and its reciprocal. Its family $\Xi_N^{(k,\theta)}$ is nonconstructive and may depend on both fixed parameters, unlike Π_N .

The completion class matters even for $k = 4$. Its standard-orientation optimum is $1/144$, but periodically reversing rows adds a local bond term to the core vector. We exploit it to construct a separate explicit family (Σ_N) with 12 or 17 insertions and discrepancy $N^3/192 + o(N^3)$. A ten-pattern functional annihilates centred additive corrections and proves optimality over all fixed-period orientations and positive limiting aspects in the stated completion framework. No constant optimum here is asserted over arbitrary permutations.

2 The every-size construction

Write $[s] = \{1, \dots, s\}$ and compare tuple-valued keys lexicographically. For even size, write $N = 2E$ and set

$$a = \lfloor \sqrt{E} \rfloor, \quad b = \left\lfloor \frac{E}{a} \right\rfloor, \quad r = E - ab.$$

Then

$$0 \leq r < a, \quad a \leq b \leq a + 2. \tag{1}$$

If $r > 0$, define

$$R = R_{b,r} = \left\{ \left\lceil \frac{jb}{r} \right\rceil : 1 \leq j \leq r \right\} \subseteq [b];$$

When $r > 0$, (1) gives $b/r > 1$, so the r ceiling values are distinct and $|R| = r$. Put $R = \emptyset$ when $r = 0$. Our cell set is

$$F_E = ([a] \times [b]) \cup (\{a+1\} \times R),$$

which has exactly E cells. The first rectangle is the *core*; the cells in the last row are *partial*. For example, $E = 7$ gives $(a, b, r) = (2, 3, 1)$, $R = \{3\}$, and hence a 2×3 core together with the single partial cell $(3, 3)$.

Place two coloured points above each cell:

$$Q_E = F_E \times \{-1, +1\}.$$

For $p = (x, y, c)$, order these $2E$ points by the two injective keys

$$K_X(p) = (x, -cy, c), \quad K_Y(p) = (y, cx, c). \tag{2}$$

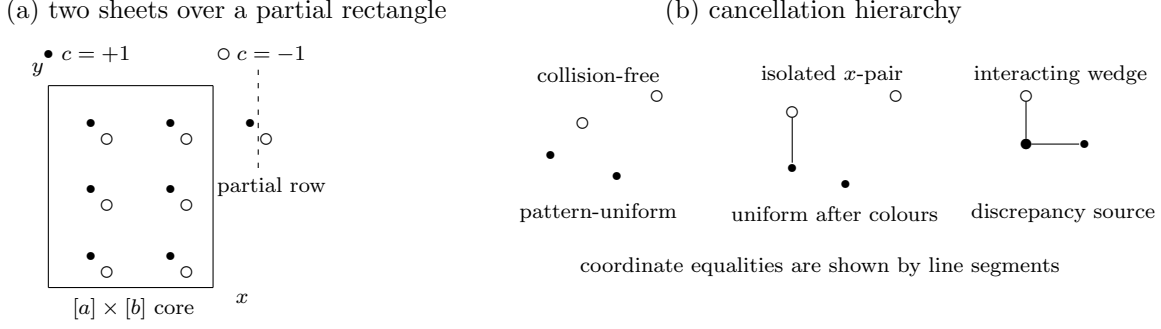


Figure 1: Construction and cancellation mechanism. (a) The case $E = 7$: a 2×3 core and one partial cell, with a positive (filled) and negative (open) point over each cell. The keys in (2) are the coordinate orders after opposite small rotations of the two sheets. (b) Collision-free classes and isolated same-axis pairs are exactly pattern-uniform after summing colours (Lemma 4.1); only interacting equalities, such as the displayed wedge, enter the discrepancy bound.

Listing the points in K_X -order and recording their ranks in K_Y -order defines $\Pi_{2E} \in \mathfrak{S}_{2E}$. More generally, for any finite cell set $F \subset \mathbb{Z}^2$, write $\Pi(F)$ for the permutation of $F \times \{-1, +1\}$ induced by these same two key orders. The final coordinate keeps both keys injective even at zero or negative coordinates; the rotation interpretation below concerns the positive-coordinate set F_E . For the preceding $E = 7$ example this gives

$$\Pi_{14} = (12, 7, 3, 2, 6, 11, 13, 8, 4, 1, 5, 10, 14, 9).$$

On the positive-coordinate set F_E , the final coordinate is redundant. Here the two orders also have an exact geometric realization. Put $M = \max\{a+1, b\}$ and choose $0 < \varepsilon < \arctan(1/(2M))$. Rotate the copy of colour c through angle $c\varepsilon$; its coordinates become

$$(x \cos \varepsilon - cy \sin \varepsilon, \quad y \cos \varepsilon + cx \sin \varepsilon).$$

A unit difference between original x -coordinates contributes $\cos \varepsilon$, while the perturbation difference has magnitude at most $2b \sin \varepsilon$; hence unequal x -coordinates keep their order. The analogous bound $2(a+1) \sin \varepsilon$ handles unequal y -coordinates. Original ties are resolved by $-cy$ and cx . Therefore the rotated coordinate orders are precisely (2). We write $\text{ES}^\pm(a, b) := \Pi([a] \times [b])$ for this complete-core permutation. It is Beniamini–Lavee–Linial’s sufficiently-small-angle order type; if $r = 0$, it is the whole construction [1, Definition 5.6].

For odd $N > 1$, obtain Π_N from Π_{N-1} by appending a point that is both rightmost and highest. Let Π_1 be the unique permutation of size one.

3 A balanced partial row

The set R distributes its r columns evenly through $[b]$.

Lemma 3.1 (prefix balance). *For every integer $0 \leq z \leq b$,*

$$|R \cap [z]| = \left\lfloor \frac{rz}{b} \right\rfloor.$$

In particular, if $d_y = \mathbf{1}_R(y) - r/b$, then

$$\left| \sum_{y=1}^z d_y \right| < 1 \quad (0 \leq z < b), \quad \sum_{y=1}^b d_y = 0.$$

Proof. For $r > 0$, the inequality $\lceil jb/r \rceil \leq z$ is equivalent to $j \leq rz/b$. Counting the eligible integers j gives the formula. The case $r = 0$ is immediate. $\square$

Lemma 3.2 (integrated mechanical discrepancy). *For $0 \leq r < b$, the mechanical row satisfies*

$$W(R) := \sum_{z=1}^{b-1} \left| |R \cap [z]| - \frac{rz}{b} \right| = \frac{b - \gcd(b, r)}{2} \leq \frac{b-1}{2}.$$

Proof. Lemma 3.1 makes the z th summand the fractional part $\{rz/b\}$. If $g = \gcd(b, r)$, the nonzero residues of rz modulo b consist of each of $g, 2g, \dots, b-g$ repeated g times. Their normalized sum is $(b-g)/2$. $\square$

For $1 \leq t \leq k$ and $1 \leq y \leq b$, use the convention that an out-of-range binomial coefficient is zero and put

$$f_t(y) = \binom{y-1}{t-1} \binom{b-y}{k-t}.$$

This counts the $(k-1)$ -subsets of $[b] \setminus \{y\}$ in which y has rank t after it is inserted.

Lemma 3.3 (rank balance). *For the mechanical row R ,*

$$\left| \sum_{y \in R} f_t(y) - \frac{r}{b} \binom{b}{k} \right| \leq \binom{b-1}{k-1}. \quad (3)$$

Proof. Vandermonde gives $\sum_y f_t(y) = \binom{b}{k}$. Put $g_y = \mathbf{1}_R(y) - r/b$ and $G_z = \sum_{y \leq z} g_y$. Lemma 3.1 gives $-1 < G_z \leq 0$ and $G_b = 0$. The nonnegative weight f_t is unimodal: where adjacent terms are positive, $f_t(y+1)/f_t(y) = y(b-y-k+t)/((y-t+1)(b-y))$ is at least one exactly when $y \leq (t-1)b/(k-1)$. Also $f_t(y) \leq \binom{b-1}{k-1}$, since it counts only the $(k-1)$ -subsets avoiding y that put y at rank t . In $\sum_y g_y f_t(y) = -\sum_{z < b} G_z (f_t(z+1) - f_t(z))$, the total increase and total decrease are each bounded by this maximum, and $0 \leq -G_z < 1$. This proves the claim, including $r = 0$. $\square$

4 Exact cancellation in the core

Consider first the coloured complete rectangle $[a] \times [b] \times \{-1, +1\}$ with the orders (2). For a k -subset S , let $r_x(S)$ and $r_y(S)$ be the numbers of distinct x - and y -coordinates, and define its *collision defect*

$$d(S) = (k - r_x(S)) + (k - r_y(S)).$$

The three configurations in Figure 1(b) explain the threshold. If all coordinates are distinct, every one of the 2^k colour words gives the same two strict coordinate orders. If exactly one x -coordinate is repeated while all ordinates are distinct, then for any prescribed pattern the coordinate assignment is forced; exactly two of the four colour pairs resolve the tied pair in the required order, and the other $k-2$ colours are free, giving 2^{k-1} realizations of every pattern. If an x -tie and a y -tie share a point, its single colour must satisfy two local resolutions, which may agree or conflict. These interacting equalities are the first classes that need an estimate.

Lemma 4.1 (same-axis pair-collision cancellation). *In the complete rectangle, fix an x -coordinate multiset and a set of distinct y -coordinates. Every pattern occurs equally often among the corresponding coloured subsets if every x -multiplicity is at most two. Symmetrically, the same holds for a fixed set of distinct x -coordinates and a fixed y -coordinate multiset whose multiplicities are at most two. In particular, the layers $d = 0$ and $d = 1$ separately have equal counts for all patterns.*

Proof. Suppose the x -multiset has q repeated pairs and all ordinates are distinct. The pairs occupy disjoint adjacent X -blocks. A prescribed pattern fixes the unordered ordinate pair in each block and the ordinate of each singleton, hence the uncoloured cell set. Reversing both colours negates the two distinct tie keys $(-cy, c)$ and reverses their comparison, so either resolution has exactly two colour choices. The q pairs use disjoint colour variables and the $k - 2q$ singleton colours are free. Thus exactly $2^q 2^{k-2q} = 2^{k-q}$ colourings realize each pattern. Exchange axes and use (cx, c) for the other assertion. Summing over coordinate choices proves uniformity; $q = 0$ and $q = 1$ give the stated defect layers. $\square$

The following transfer statement does not require a rectangle or positive coordinates. For a finite cell set $F \subset \mathbb{Z}^2$, put $E = |F|$. Write r_x and c_y for its row and column degrees, and set

$$R_3 = \sum_x r_x^3, \quad C_3 = \sum_y c_y^3, \quad M_{11} = \sum_{(x,y) \in F} r_x c_y.$$

For a vector indexed by $\mathfrak{S}_k$, let $\|\cdot\|_{\text{cen}}$ denote maximum distance from its coordinate average.

Lemma 4.2 (independent-jitter transfer). *Let P_F be the k -profile vector of $\Pi(F)$. For each ordered cell tuple $z \in F^k$, independently refine every equality class of its x - and y -coordinates by a uniform label order, and let $q_z(\tau)$ be the probability that the two refined orders induce τ . Define the unnormalized jitter-mass vector by*

$$J_F(\tau) = 2^k \sum_{z \in F^k} q_z(\tau).$$

Then

$$\|k!P_F - J_F\|_{\text{cen}} \leq 2^k I_k(F), \quad (4.1)$$

where, for $k \geq 3$,

$$I_k(F) = \binom{k}{3} (R_3 + C_3) E^{k-3} + k(k-1)(k-2) M_{11} E^{k-3} + \binom{k}{2} E^{k-1}. \quad (4.2)$$

For $k = 2$, the right side of (4.1) is interpreted with $I_2(F) = E$.

Proof. For $z = (z_1, \dots, z_k) \in F^k$ and $c = (c_1, \dots, c_k) \in \{-1, +1\}^k$, let $\text{pat}(z, c)$ be the pattern induced by the K_X - and K_Y -orders of the labelled signed points $p_i = (z_i, c_i)$ when they are pairwise distinct.

$$A_z(\tau) = \sum_{c \in \{-1, +1\}^k} \mathbf{1}\{p_1, \dots, p_k \text{ are distinct and } \text{pat}(z, c) = \tau\}, \quad B_z(\tau) = 2^k q_z(\tau).$$

Every coloured k -subset is represented by its $k!$ label orders, while q_z is a probability vector. Thus, componentwise,

$$k!P_F = \sum_{z \in F^k} A_z, \quad J_F = \sum_{z \in F^k} B_z.$$

Join two labels when their cells have the same x -coordinate or the same y -coordinate. Call z good when this union graph is a matching and no two labels occupy the same cell. If the graph has m edges, every choice of its m independent jitter resolutions has probability 2^{-m} . We spell out the corresponding colour count. On an x -edge $\{i, j\}$ the ordinates are distinct, so the relative Y -order is already fixed. The tie keys are $u_i = (-c_i y_i, c_i)$ and $u_j = (-c_j y_j, c_j)$. They are distinct: equality would force $c_i = c_j$ and then $y_i = y_j$. The fixed-point-free involution $(c_i, c_j) \mapsto (-c_i, -c_j)$ sends (u_i, u_j) to $(-u_i, -u_j)$, reversing their strict lexicographic comparison. It therefore pairs each resolution with its opposite, so exactly two of the four colour pairs realize either order. This argument includes negative and zero ordinates and ties in the first key component. On a y -edge, the distinct abscissae give the identical argument for $(c_i x_i, c_i)$ and $(c_j x_j, c_j)$. Because the union graph is a matching, these edge constraints use disjoint colour variables; because repeated cells are excluded, no edge is simultaneously an x - and a y -edge. The $k - 2m$ singleton colours are free, so each joint resolution has $2^m 2^{k-2m} = 2^{k-m}$ realizing words. The 2^m resolutions partition all 2^k colour words. Each has jitter probability 2^{-m} and hence mass 2^{k-m} in $B_z = 2^k q_z$ as well. Thus $A_z = B_z$ for every good z .

Disagreement can therefore occur only when an axis contains three chosen labels, one label is incident to an x -edge and a y -edge with different neighbours, or a label pair occupies the same cell. For the first event, choose the three responsible labels and the common row (or column), then choose their cells and the remaining cells; this gives $\binom{k}{3}(R_3 + C_3)E^{k-3}$. For a mixed wedge, choose its ordered centre and two neighbours and then its centre cell; the two incident choices give $r_x c_y$, hence $k(k-1)(k-2)M_{11}E^{k-3}$. A repeated cell is bounded by choosing its label pair, the cell, and the remaining cells, giving $\binom{k}{2}E^{k-1}$. These are union bounds, so overlaps need not be subtracted. For $k = 2$, only the repeated-cell event remains.

Finally, if nonnegative vectors $u, v \in \mathbb{R}^n$ have coordinate sums $U, V \leq M$, then, coordinate by coordinate,

$$\left| (u_i - v_i) - \frac{U - V}{n} \right| \leq \max \left\{ U \left(1 - \frac{1}{n} \right) + \frac{V}{n}, V \left(1 - \frac{1}{n} \right) + \frac{U}{n} \right\} \leq M.$$

For every bad tuple, A_z and B_z are nonnegative vectors with coordinate sums at most 2^k (the sum for B_z is exactly 2^k). Taking $n = k!$ and $M = 2^k$, then summing over the bad-tuple union bound, proves (4.1). $\square$

For the complete $a \times b$ rectangle, summing over all ordered cell tuples is equivalent to choosing the labelled x -coordinates independently and uniformly from $[a]$, and the labelled y -coordinates independently and uniformly from $[b]$. Randomly refining ties therefore makes each global order uniform; independence of the coordinate choices and refinements makes the two orders independent. Hence J_F is exactly uniform. With $A = ab$ and $\eta = b/a$, Lemma 4.2 therefore gives

$$\delta_{\Pi([a] \times [b]), k} \leq \frac{2^k}{k!} A^{k-1} K_{k, \eta}, \quad K_{k, \eta} = \binom{k}{3}(\eta + \eta^{-1}) + \binom{k}{2}(2k - 3). \quad (4.3)$$

5 Balanced boundary completion

We now count subsets that use the partial row. A subset with at least two partial-row points belongs to a family of size at most

$$\binom{2r}{2} \binom{2E}{k-2} \leq \frac{2^{k-1}}{(k-2)!} E^{k-1}. \quad (4)$$

Consider a subset with exactly one partial-row point. In this count a *coordinate collision* means equality of original x - or y -coordinates among selected signed points. The two opposite-colour points over one cell are distinct labelled points, but selecting both creates a collision on each axis; selecting the same labelled point twice is never allowed. For $k \geq 3$, a labelled $(k-1)$ -tuple with a repeated coordinate realizes one of $J_k = (k-1)(k-2)$ elementary equality events. For a fixed event there are at most bA^{k-2} core cell assignments, since $a \leq b$. Dividing by $(k-1)!$, choosing colours and one of the $2r$ partial points, and using $rb \leq ab = A$, gives

$$\frac{2^k J_k}{(k-1)!} A^{k-1}.$$

This single event count includes repetitions on either axis. For $k = 2$ this repeated-coordinate subclass is empty. There is one additional collision possibility: a core point may share the partial point's y -coordinate. A union bound gives at most

$$(2r)(2a) \binom{2ab}{k-2} \leq \frac{2^k}{(k-2)!} E^{k-1} \quad (5)$$

subsets of this kind. The partial row has a new x -coordinate, so there is no analogous shared- x case.

It remains to analyze subsets with exactly one partial-row point and no coordinate collision. Their exact number is

$$2^k r(k-1)! \binom{a}{k-1} \binom{b-1}{k-1}.$$

Indeed, choose the signed partial point, the distinct core abscissae, the core ordinates other than its ordinate, their bijection, and the 2^{k-1} core colours. Thus a crude cardinality bound is insufficient: when $r \asymp a \asymp b \asymp E^{1/2}$, this class has $\Theta_k(E^{k-1/2})$ members. The next lemma uses the bounded rank discrepancy of the mechanical row to show that its *centred profile*, rather than its size, is only $O_k(E^{k-1})$.

For $k \geq 2$, define $m_k = 1$ when $k = 2$ and $m_k = 2$ when $k \geq 3$.

Lemma 5.1 (clean boundary profile). *Let A_τ be the number of collision-free k -subsets that contain exactly one partial-row point and induce τ . If $t = \tau(k)$, then*

$$A_\tau = 2^k \binom{a}{k-1} \sum_{y \in R} \binom{y-1}{t-1} \binom{b-y}{k-t}. \quad (6)$$

Consequently,

$$\left| A_\tau - \frac{1}{k!} \sum_{\sigma \in \mathfrak{S}_k} A_\sigma \right| \leq 2^k m_k \binom{a}{k-1} \binom{b-1}{k-1} \leq \frac{2^k m_k}{((k-1)!)^2} E^{k-1}. \quad (7)$$

Proof. For $k = 2$ the argument below has one core point and gives the same formula directly, with $m_2 = 1$. We therefore treat all $k \geq 2$ together. The partial row is last in X -order. Fix its ordinate $y \in R$. For it to have rank t among the selected ordinates, choose $t-1$ ordinates below y and $k-t$ above it. Choose the other $k-1$ distinct abscissae from the a core rows. Once these coordinate sets and a pattern τ with $\tau(k) = t$ are fixed, τ specifies a unique bijection from the core abscissae to the non-special ordinates. There are no ties, so all 2^k colourings induce the same pattern. This proves (6).

Write the sum in Lemma 3.3 as $r \binom{b}{k}/b + e_t$, where $|e_t| \leq \binom{b-1}{k-1} \leq m_k \binom{b-1}{k-1}$. Moreover, $\sum_t f_t(y) = \binom{b-1}{k-1}$, and Vandermonde's identity also gives $k \binom{b}{k}/b = \binom{b-1}{k-1}$. Hence $\sum_{t=1}^k e_t = 0$. Each final rank occurs for exactly $(k-1)!$ patterns, so the class average has zero error term. Multiplying the bound on e_t by $2^k \binom{a}{k-1}$ proves (7). $\square$

Put

$$\beta_k = 1 - \frac{1}{k!}, \quad J_k = (k-1)(k-2),$$

and, for $\rho \geq 1$, define

$$\begin{aligned} C_\rho(k) &= \frac{2m_k}{((k-1)!)^2} + \frac{2K_{k,\rho}}{k!} \\ &\quad + \beta_k \left(\frac{2J_k}{(k-1)!} + \frac{3}{(k-2)!} \right) + \frac{\beta_k}{(k-1)!}, \\ K_{k,\rho} &= \binom{k}{3}(\rho + \rho^{-1}) + \binom{k}{2}(2k-3). \end{aligned} \tag{8}$$

The last term is reserved for odd padding. The binomial convention makes every displayed class count valid even when there are too few rows or columns; in particular, the complete-core class is simply empty when $k > 2ab$.

Proof of Theorem 1.1 for even N . If $k > 2ab$, the class using only the $2ab$ core points is empty; throughout this proof its centred contribution is then interpreted as zero. Otherwise the complete core contributes discrepancy at most (4.3). The clean one-partial-point class contributes at most (7). Put every remaining partial-row subset into $\mathcal{B}$. Equations (4)–(5) give, with $\rho_E = b/a$,

$$|\mathcal{B}| \leq 2^{k-1} \left(\frac{2J_k}{(k-1)!} + \frac{3}{(k-2)!} \right) E^{k-1}.$$

Since the centered contribution of an arbitrary class of size L is at most $\beta_k L$, and $N = 2E$, the three contributions give

$$\overline{C}_E(k) = C_{\rho_E}(k) - \mathbf{1}_{\{k > 2ab\}} \frac{2K_{k,\rho_E}}{k!}$$

and

$$\delta_{\Pi_N,k} \leq \left(\overline{C}_E(k) - \frac{\beta_k}{(k-1)!} \right) N^{k-1}. \tag{9}$$

To normalize (8), use $\beta_k \leq 1$ and set $\eta = \rho + \rho^{-1}$. The clean term is at most $2/k^2$ after multiplication by $k!/k^3$; the bad and reserved terms sum to $(2J_k + 3(k-1) + 1)/k^2 = 2 - 3/k + 2/k^2$. If $k \leq 2ab$, then $b - a \leq 2$ gives $\eta - 2 = (b - a)^2/(ab) \leq 8/k$, and therefore

$$\begin{aligned} \frac{k!}{k^3} C_{\rho_E}(k) &\leq \frac{\eta(k-1)(k-2)}{3k^2} + \frac{(k-1)(2k-3)}{k^2} + 2 - \frac{3}{k} + \frac{4}{k^2} \\ &\leq \frac{14k^3 - 22k^2 + k + 16}{3k^3} < \frac{14}{3}. \end{aligned}$$

If $k > 2ab$, omit the empty core term. The remaining normalized sum is at most $(2k^2 - 3k + 4)/k^2 < 14/3$. Thus $\overline{C}_E(k) < 14k^3/(3k!)$ in both cases. $\square$

6 Odd orders and completion of the proof

Lemma 6.1 (one-point padding). *Let $2 \leq k \leq M + 1$. Suppose $\pi \in \mathfrak{S}_M$ and $\pi^+ \in \mathfrak{S}_{M+1}$ is obtained by appending a rightmost-and-highest point. Write*

$$D_k(\tau) = \#_\tau(\pi) - \frac{1}{k!} \binom{M}{k}, \quad D_k^+(\tau) = \#_\tau(\pi^+) - \frac{1}{k!} \binom{M+1}{k}, \quad c_{M,k} = \frac{1}{k!} \binom{M}{k-1}.$$

Set $D_{M+1}(\tau) = 0$, since π has no $(M+1)$ -subsets, and $D_1(1) = 0$. If $\tau(k) = k$, let τ' be obtained by deleting its last entry. Then

$$D_k^+(\tau) = D_k(\tau) + \begin{cases} D_{k-1}(\tau') + (k-1)c_{M,k}, & \tau(k) = k, \\ -c_{M,k}, & \tau(k) \neq k. \end{cases} \quad (10)$$

With the conventions $\delta_{\pi,1} = \delta_{\pi,M+1} = 0$,

$$\delta_{\pi^+,k} \leq \delta_{\pi,k} + \beta_k \binom{M}{k-1}, \quad (11)$$

$$\delta_{\pi^+,k} \leq \delta_{\pi,k} + \delta_{\pi,k-1} + \frac{k-1}{k!} \binom{M}{k-1}. \quad (12)$$

Proof. The number X_τ of new copies is $\#_{\tau'}(\pi)$ when $\tau(k) = k$, and zero otherwise. Pascal's identity gives (10). Centering the range $0 \leq X_\tau \leq \binom{M}{k-1}$ at $c_{M,k}$ gives (11); taking absolute values in the recurrence gives (12). $\square$

For odd $N \geq k+1$, let $M = N-1$. Combining (11) with the reserved term in the even estimate gives the explicit inequality

$$\delta_{\Pi_N,k} \leq \left(\overline{C}_E(k) - \frac{\beta_k}{(k-1)!} \right) M^{k-1} + \beta_k \binom{M}{k-1} \leq \overline{C}_E(k) M^{k-1} \leq \overline{C}_E(k) N^{k-1}.$$

If $N = k$ is odd, then $\delta_{\Pi_k,k} = \beta_k < 1$, whereas $14k^{k+2}/(3k!) > 1$ for $k \geq 3$. Thus the same strict global bound holds. This proves the bound in Theorem 1.1(i); its fixed- k exponent claim follows from Beniamini–Lavee–Linial's lower bound [1, Theorem 7].

Corollary 6.2 (absolute-density consequences). *Uniformly over $\tau \in \mathfrak{S}_k$ and $2 \leq k = o(N^{1/2})$,*

$$\left| \frac{\#_\tau(\Pi_N)}{\binom{N}{k}} - \frac{1}{k!} \right| \leq \left(\frac{14}{3} + O\left(\frac{k^2}{N}\right) \right) \frac{k^3}{N}, \quad (13)$$

with an absolute implied constant. For $N \geq 2k^2 + 1$, the left side is less than $7k^3/N$.

Proof. The product $f_{N,k} = \prod_{j < k} (1 - j/N)$ is $1 + O(k^2/N)$ in the stated range. Divide Theorem 1.1(i) by $\binom{N}{k} = (N^k/k!)f_{N,k}$. For the finite bound use $f_{N,k} \geq 1 - k(k-1)/(2N) > 3/4$ and $56/9 < 7$. $\square$

7 Exact rectangular profiles and consequences

The exact binomial-basis formula determines the growing-pattern threshold and then the local leading vector for fixed k .

7.1 An exact formula in the binomial basis

The preceding defect argument can be sharpened to an exact formula. Write $\alpha \models k$ for a positive composition of k , let $\ell(\alpha)$ be its number of parts, and let $p_\alpha(i)$ be the index of the part containing position i . Thus a composition records consecutive equality blocks in an ordered list.

Throughout this section the coordinate sets are $[a]$ and $[b]$, so both coordinates are positive. For $\tau \in \mathfrak{S}_k$, compositions $\alpha, \beta \models k$, and $c \in \{-1, +1\}^k$, put $p_i = (p_\alpha(i), p_\beta(\tau(i)), c_i)$, with block indices starting at one. Let $w_\tau(\alpha, \beta)$ count the colour words for which $p_1, \dots, p_k$ are strictly increasing in K_X and $p_{\tau^{-1}(1)}, \dots, p_{\tau^{-1}(k)}$ are strictly increasing in K_Y . Only comparisons within equal-coordinate blocks require checking.

For example, if $k = 2$, $\alpha = (2)$, $\beta = (1, 1)$, and $\tau = 12$, the Y -blocks are singletons and the admissible words are $(+, -)$ and $(-, -)$; hence $w_{12}((2), (1, 1)) = 2$.

Theorem 7.1 (exact rectangular profile). *For every $a, b \geq 1$, every $2 \leq k \leq 2ab$, and every $\tau \in \mathfrak{S}_k$,*

$$\#_\tau(\text{ES}^\pm(a, b)) = \sum_{\alpha, \beta \models k} w_\tau(\alpha, \beta) \binom{a}{\ell(\alpha)} \binom{b}{\ell(\beta)}. \quad (14)$$

Thus every rectangular profile entry is an explicit bivariate polynomial in a and b .

Proof. List a selected set as $p_1, \dots, p_k$ in final X -order. Its equal original x -coordinates form consecutive blocks, giving a unique composition α ; the equal original y -coordinates in final Y -order similarly give β . If the induced pattern is τ , then p_i lies in original-coordinate blocks $(p_\alpha(i), p_\beta(\tau(i)))$.

Choose the distinct original coordinate values in $\binom{a}{\ell(\alpha)} \binom{b}{\ell(\beta)}$ ways. Within an equal- x block, the key $-cy$ orders all positive colours with decreasing y -block index, followed by all negative colours with increasing index; this is exactly the required K_X order. The key cx gives the required K_Y order symmetrically. Strictness also excludes selecting the same coloured point twice. Hence admissible colour words are in bijection with the selected sets having these data, proving (14). $\square$

7.2 The sharp growing-pattern range

The target $1/k!$ is too small for an absolute collision-probability bound to give relative balance. We instead retain the exact cancellation of disjoint isolated pairs before estimating connected interactions.

Lemma 7.2 (connected collision bound). *Put $s = \min(a, b)$ and $n = 2ab$. For $2 \leq k$ and $64k < s$, uniformly over $\tau \in \mathfrak{S}_k$,*

$$\left| \frac{k! \#_\tau(\text{ES}^\pm(a, b))}{\binom{n}{k}} - 1 \right| \leq \exp \left(\frac{4096k^3}{s^2(1 - 64k/s)} + \frac{k(k-1)}{2(n-k+1)} \right) - 1. \quad (15)$$

Proof. Fix τ and set $\rho = \tau^{-1}$. Form a multigraph on $[k]$ with X -edges $(i, i+1)$ and Y -edges $(\rho(j), \rho(j+1))$, retaining parallel edges. A selected edge set S specifies equality compositions α, β on the two axes. Put

$$F(S) = \prod_i \alpha_i! \prod_j \beta_j!, \quad W(S) = 2^{-k} w_\tau(\alpha, \beta) F(S).$$

Let $\mathbf{S}$ be the equality mask of two independent sorted samples of k uniform variables from $[a]$ and $[b]$. A composition α on the first axis has probability $\binom{a}{\ell(\alpha)} k! / (a^k \prod_i \alpha_i!)$. Thus (14) gives the exact identity

$$\mathbb{E}W(\mathbf{S}) = \frac{(k!)^2 \#_{\tau}(\text{ES}^{\pm}(a, b))}{n^k}. \quad (16)$$

The weight $W(S)$ factors over the connected components of S . Each constraint uses the end-point colours and the relative order or equality of their opposite-coordinate blocks. That equality forces the connecting monochromatic path into the same component, so changing blocks outside it does not alter the constraint. This also covers repeated cells; parallel edges are one possible connected interaction, not independent factors. Isolated vertices have weight one. Moreover $W(\emptyset) = W(\{e\}) = 1$, by the two-versus-two tie count.

Define $c(T) = \sum_{U \subseteq T} (-1)^{|T|-|U|} W(U)$, so $W(S) = \sum_{T \subseteq S} c(T)$. Factoring the Boolean sum over components shows that $c(T)$ is their coefficient product. A component consisting of one edge therefore makes $c(T) = 0$. For a component with d edges, $0 \leq W(U) \leq F(U) \leq F(T)$ for every $U \subseteq T$: splitting a block can only decrease its factorial product. Consequently $|c(T)| \leq 2^d F(T)$, also after multiplying over components.

To control the equality events jointly, realize each discrete sample by $1 + \lfloor aU_i \rfloor$ with independent continuous uniform U_i . An X -run of r vertices forces its $r - 1$ intervening order-statistic gaps to have sum at most $1/a$. Distinct runs use disjoint gaps. The joint density of any d_X gaps is $(k)_{d_X} (1 - \sum z_i)^{k-d_X} \leq (k)_{d_X}$ on its simplex. Hence

$$\mathbb{P}(T_X \subseteq \mathbf{S}) \leq \frac{(k)_{d_X} a^{-d_X}}{\prod_{X\text{-runs}} (r-1)!},$$

where $(k)_d = k!/(k-d)!$. This is a joint bound, not an independence assumption between runs. Applying it independently to the two axes, and using $r!/(r-1)! = r \leq 2^{r-1}$, gives for every T

$$|c(T)| \mathbb{P}(T \subseteq \mathbf{S}) \leq (4k/a)^{d_X} (4k/b)^{d_Y}. \quad (17)$$

The multigraph has maximum degree four. There are at most $k16^d$ connected d -edge sets: choose a root and encode a tour of length $2d$ traversing each edge in both directions, with at most four choices per step. Let $\mathcal{C}$ contain all connected edge sets of size at least two and put $q(C) = (4k/a)^{d_X(C)} (4k/b)^{d_Y(C)}$. Expanding into components, applying (17), and then dropping the disjointness restriction on component collections yields

$$|\mathbb{E}W - 1| \leq \prod_{C \in \mathcal{C}} (1 + q(C)) - 1 \leq \exp \left(k \sum_{d \geq 2} (64k/s)^d \right) - 1.$$

Finally, if $f_{n,k} = (n)_k/n^k$, then $\log f_{n,k}^{-1} \leq k(k-1)/(2(n-k+1))$. Divide (16) by $f_{n,k}$ and use $|\mathbb{E}W/f_{n,k} - 1| \leq e^{\sum_C q(C)}/f_{n,k} - 1$ to obtain (15). $\square$

Corollary 7.3 (relative balance for growing patterns). *If $2 \leq k = k(N) = o(N^{1/3})$, then, uniformly over $\tau \in \mathfrak{S}_k$,*

$$\frac{k! \#_{\tau}(\Pi_N)}{\binom{N}{k}} = 1 + O \left(\frac{k}{\sqrt{N}} + \frac{k^3}{N} \right),$$

with an absolute implied constant.

Proof. For $a = \lfloor \sqrt{\lfloor N/2 \rfloor} \rfloor$, the construction gives

$$\text{ES}^\pm(a, a) \preceq \Pi_N \preceq \text{ES}^\pm(a+3, a+3), \quad (18)$$

where $\preceq$ denotes induced containment. At odd sizes the northeast point is represented by $(a+2, b+1, +1)$, whose two key ranks are last. Both square sizes differ from N by $O(\sqrt{N})$. In the stated range, their binomial normalizations differ from $\binom{N}{k}$ by factors $\exp(O(k/\sqrt{N}))$, while Lemma 7.2 gives relative error $O(k^3/N)$ on either square. Monotonicity of pattern counts in (18) proves the assertion. $\square$

Proposition 7.4 (increasing patterns at the critical scale). *For positive integers a, b and $1 \leq k \leq 2ab$, write $I_{a,b}(k) = \#_{12\dots k}(\text{ES}^\pm(a, b))$. Then*

$$I_{a,b}(k) = 2 \sum_{j=0}^{k-1} \binom{k-1}{j} \binom{a+k-1-j}{k} \binom{b+j}{k}, \quad (19)$$

where a binomial coefficient is zero when its upper argument is smaller than its lower argument. In particular, the longest increasing subsequence has length $a+b-1$. For the universal family and $2 \leq k = k(N) \leq N$, if $k/N^{1/3} \rightarrow c \in [0, \infty)$, then

$$\frac{k! \#_{12\dots k}(\Pi_N)}{\binom{N}{k}} \rightarrow e^{-c^3/6}. \quad (20)$$

If $k/N^{1/3} \rightarrow \infty$, the same relative density tends to zero.

Proof. An increasing occurrence is a chain of distinct cells with both coordinates nondecreasing. A repeated cell is impossible because its two colours have opposite X - and Y -orders. Comparing the keys shows that a $+$ colour at the next point requires a strict increase in x , whereas a $-$ colour requires a strict increase in y ; these are the only conditions. A colour word with j plus signs after its first position therefore has $\binom{a+k-1-j}{k} \binom{b+j}{k}$ possible cell chains. Indeed, a weakly increasing k -tuple in $[a]$ with j specified strict gaps is counted by $\binom{a+k-1-j}{k}$, by subtracting the number of preceding strict gaps. Summing over the $2^{\binom{k-1}{j}}$ colour words proves (19). Its nonzero terms satisfy $\max(0, k-b) \leq j \leq \min(k-1, a-1)$, giving the chain-length assertion.

For a square let $J \sim \text{Bin}(k-1, 1/2)$ and $D_{m,k} = \prod_{r=0}^{k-1} (1 - r/(2m^2))$. Reversing one falling factorial in (19) gives

$$\frac{k! I_{m,m}(k)}{\binom{2m^2}{k}} = \frac{1}{D_{m,k}} \mathbb{E} \prod_{r=0}^{k-1} \left(1 - \frac{(J-r)^2}{m^2} \right). \quad (21)$$

This identity holds throughout $1 \leq k \leq 2m^2$. Terms outside $\max(0, k-m) \leq J \leq \min(k-1, m-1)$ vanish; all factors are positive on this support. If $k/(2m^2)^{1/3} \rightarrow c < \infty$, then $k = o(m)$ and

$$\sum_{r=0}^{k-1} (J-r)^2 = k \left(J - \frac{k-1}{2} \right)^2 + \frac{k(k^2-1)}{12}.$$

After division by m^2 , the first term has expectation $k(k-1)/(4m^2) = o(1)$ and the second tends to $c^3/6$. The uniform logarithmic remainder is $O(k^5/m^4) = o(1)$, while $\log D_{m,k} = O(k^2/m^2) = o(1)$. The products lie in $[0, 1]$, so convergence in probability also gives convergence of expectations. This proves the critical limit on squares, including $c = 0$.

For $2 \leq k \leq 2m - 1$, bounding the product in (21) by its quadratic exponential and using $D_{m,k}^{-1} \leq \exp(k(k-1)/(2m^2))$ gives

$$\frac{k! I_{m,m}(k)}{\binom{2m^2}{k}} \leq \exp\left(-\frac{k(k-1)(k-5)}{12m^2}\right).$$

For $k \geq 2m$ the count is zero. Thus the relative increasing density tends to zero whenever $k^3/m^2 \rightarrow \infty$.

Finally use (18). For $k = O(\sqrt{N})$, its square binomial normalizations differ from $\binom{N}{k}$ by $\exp(O(k/\sqrt{N}))$. At the critical scale this factor tends to one, and the square limits squeeze the middle count. In the supercritical regime the upper-square count is zero for $k \geq 2(a+3)$; otherwise the normalization factor is bounded and the square exponential tends to zero. Together with Corollary 7.3, this proves the equivalence in Theorem 1.1(ii): if $k/N^{1/3}$ does not tend to zero, take a subsequence converging to a positive finite value or to infinity. The increasing pattern then prevents relative balance. $\square$

7.3 The local defect-two formula

For fixed k , the first nonuniform terms have coordinate defect two. The following local statistics encode their colour sums. Put $\rho = \tau^{-1}$ and define

$$\begin{aligned} T_x(\tau) &= \#\{i \leq k-2 : \tau(i+1) \neq \max(\tau(i), \tau(i+1), \tau(i+2))\}, \\ T_y(\tau) &= T_x(\tau^{-1}), \quad Q_k = \binom{k-2}{2}, \\ H_i &= \mathbf{1}_{i < k, \tau(i+1) < \tau(i)} - \mathbf{1}_{i > 1, \tau(i-1) < \tau(i)}, \\ V_i &= \mathbf{1}_{\tau(i) < k, \rho(\tau(i)+1) < i} - \mathbf{1}_{\tau(i) > 1, \rho(\tau(i)-1) < i}, \\ M(\tau) &= (k-1)^2 - \sum_{i=1}^k H_i V_i. \end{aligned}$$

Here and below a false subscript condition contributes zero.

Lemma 7.5 (explicit defect two). *Let $D_\tau^{(2)}(a, b)$ count core subsets of coordinate defect exactly 2. Then*

$$\begin{aligned} D_\tau^{(2)}(a, b) &= 2^{k-2} \left[(Q_k + T_x(\tau)) \binom{a}{k-2} \binom{b}{k} \right. \\ &\quad + M(\tau) \binom{a}{k-1} \binom{b}{k-1} \\ &\quad \left. + (Q_k + T_y(\tau)) \binom{a}{k} \binom{b}{k-2} \right]. \end{aligned}$$

Proof. Writing $d_x = k - r_x$ and $d_y = k - r_y$, defect two means $d_x + d_y = 2$. Thus (d_x, d_y) is $(2, 0)$, $(1, 1)$, or $(0, 2)$; a defect-two equality partition on one axis is either one triple or two disjoint pairs. After factoring out 2^{k-2} compatible colour choices, the five defect-two types and their normalized

contributions are recorded in the next display.

type	normalized placements	coordinate choices
two disjoint x -pairs	Q_k	$\binom{a}{k-2} \binom{b}{k}$
one x -triple	$T_x(\tau)$	$\binom{a}{k-2} \binom{b}{k}$
one pair on each axis	$M(\tau)$	$\binom{a}{k-1} \binom{b}{k-1}$
two disjoint y -pairs	Q_k	$\binom{a}{k} \binom{b}{k-2}$
one y -triple	$T_y(\tau)$	$\binom{a}{k} \binom{b}{k-2}$

Indeed, two disjoint x -pairs occupy two nonadjacent edges of the position path, giving $Q_k = \binom{k-2}{2}$. For an x -triple, the colours form a positive prefix with decreasing ordinates followed by a negative suffix with increasing ordinates. If the middle ordinate is largest, no cut works; otherwise exactly two cuts work. This gives the T_x row, and inversion gives the two y -rows.

It remains to verify the mixed row. Let $e = (u, u+1)$ be an adjacent position edge and let $f = (s, s+1)$ be an adjacent value edge, pulled back to positions $\rho(s), \rho(s+1)$. Write their active-endpoint functions as

$$h_e(i) = \mathbf{1}_{\{i=u, \tau(u) > \tau(u+1)\}} - \mathbf{1}_{\{i=u+1, \tau(u) < \tau(u+1)\}},$$

$$v_f(i) = \mathbf{1}_{\{i=\rho(s), \rho(s) > \rho(s+1)\}} - \mathbf{1}_{\{i=\rho(s+1), \rho(s+1) > \rho(s)\}}.$$

First suppose that e and the pulled-back f have the same two endpoints. The two selected labels then lie over one cell, so distinctness forces their colours to be opposite. Comparing the two key orders shows that the $+$ point comes first in X and second in Y . Consequently the induced pair is necessarily a descent: the normalized weight is one for the descent and zero for the ascent. In the descent the two active endpoints below are different, so $\sum_i h_e(i) v_f(i) = 0$; in the ascent they coincide with product one. Thus the desired weight $1 - \sum_i h_e(i) v_f(i)$ already holds in this repeated-cell case.

Assume now that the two equality edges do not have the same pair of endpoints. Each equality imposes exactly one colour constraint. If $\tau(u) > \tau(u+1)$, the two points on e appear in the required X -order exactly when the left endpoint has colour $+$; if $\tau(u) < \tau(u+1)$, exactly when the right endpoint has colour $-$. Thus the active endpoint of e and its forced colour are encoded by the unique nonzero value $h_e(i) \in \{1, -1\}$. For f , the active endpoint is the later of $\rho(s), \rho(s+1)$, and its forced colour is $-v_f(i)$. These assertions follow immediately by comparing the tied keys $-cy$ and cx , respectively.

If the two active endpoints are different, the constraints concern two colours, so there are 2^{k-2} compatible words and the normalized weight is one. If they coincide at i , two cases remain:

relation	two constraints	normalized compatible-word count
$h_e(i) = -v_f(i)$	agree	2
$h_e(i) = v_f(i)$	conflict	0.

Both cases are $1 - h_e(i) v_f(i)$. Because each edge has only one active endpoint, summing over all common endpoints gives the normalized weight

$$1 - \sum_i h_e(i) v_f(i)$$

for every noncoincident pair (e, f) . Together with the repeated-cell calculation above, this proves the formula for every edge pair. Summing over all $(k-1)^2$ pairs gives

$$(k-1)^2 - \sum_i \left(\sum_{e \ni i} h_e(i) \right) \left(\sum_{f \ni i} v_f(i) \right) = (k-1)^2 - \sum_i H_i V_i = M(\tau).$$

The coordinate choices in the table now prove the lemma. $\square$

Define the centered local vector and its norm by

$$g_k(\tau) = T_x(\tau) + T_y(\tau) + \frac{11-4k}{3} - \frac{k}{k-1} \sum_{i=1}^k H_i V_i, \quad (22)$$

$$\lambda_{k,\tau} = \frac{2^{k-2}}{k!(k-2)!} g_k(\tau), \quad \Lambda_k = \frac{2^{k-2}}{k!(k-2)!} \max_{\tau} |g_k(\tau)|. \quad (23)$$

For each fixed k ,

$$\#_{\tau}(\text{ES}^{\pm}(m, m)) - \frac{1}{k!} \binom{2m^2}{k} = \lambda_{k,\tau} m^{2k-2} + O_k(m^{2k-3}), \quad (24)$$

$$\delta_{\text{ES}^{\pm}(m, m), k} = \Lambda_k m^{2k-2} + O_k(m^{2k-3}). \quad (25)$$

These statistics compress the two one-axis triple classes and the mixed pair class in the preceding table. For example, $\tau = 4213$ has $T_x = 2$, $T_y = 1$, and $\sum_i H_i V_i = 0$, so $g_4(4213) = 4/3$ and $\lambda_{4,4213} = 1/9$. In the square case, the coefficient in Lemma 7.5, after factoring out $2^{k-2}/(k!(k-2)!)$, is

$$R_k(\tau) = T_x(\tau) + T_y(\tau) + 2Q_k + \frac{k}{k-1} M(\tau).$$

For uniform $\tau \in \mathfrak{S}_k$, write $\gamma_k = 2^{k-2}/(k!(k-2)!)$. The defect-zero and defect-one classes are pattern-uniform and together contribute

$$P_{01}(m) = 2^k \left[\binom{m}{k}^2 + (k-1) \binom{m}{k} \binom{m}{k-1} \right].$$

There are $k-1$ possible doubled blocks on each axis, each with colour weight 2^{k-1} , which gives the second term. Expanding the falling factorials, the coefficient of m^{2k-2} in P_{01} is $\gamma_k(-6k^2 + 14k - 10)/3$. The profile average is $\binom{2m^2}{k}/k!$, whose corresponding coefficient is $-\gamma_k$. Since the defect-two contribution is $\gamma_k R_k(\tau)$, it follows that

$$\mathbb{E}R_k = \frac{6k^2 - 14k + 7}{3}, \quad R_k(\tau) - \mathbb{E}R_k = T_x + T_y + \frac{11-4k}{3} - \frac{k}{k-1} \sum_i H_i V_i.$$

This identifies the centered defect-two coefficient as $\lambda_{k,\tau}$. All higher defects have degree at most $2k-3$, proving (24); taking the maximum over the fixed finite set $\mathfrak{S}_k$ gives (25). Also $\mathbb{E}T_x = \mathbb{E}T_y = 2(k-2)/3$, since each triple has a nonmaximal middle entry with probability $2/3$. Substitution in R_k gives $\mathbb{E} \sum_i H_i V_i = (k-1)/k$ and hence $\mathbb{E}M = (k-1)^2 - (k-1)/k$, including $k=2$. After forming τ^{-1} , each $g_k(\tau)$ costs $O(k)$ to evaluate, so Λ_k is computable in $O(k k!)$ time.

7.4 The fixed-aspect formula and a four-point baseline

The exact formula and the defect-two lemma also give the following unoptimized fixed-aspect consequence. Put $\mu_k = 2(k-2)/3$ and $S(\tau) = \sum_i H_i V_i$. For $\theta > 0$, define

$$h_{k,\tau}(\theta) = \theta^{-1}(T_x(\tau) - \mu_k) + 1 - \frac{k}{k-1} S(\tau) + \theta(T_y(\tau) - \mu_k). \quad (26)$$

If $a, b \rightarrow \infty$ with $a/b \rightarrow \theta$, the defect-two terms in Lemma 7.5, after centering by the uniform profile, give

$$\#_\tau(\text{ES}^\pm(a, b)) - \frac{1}{k!} \binom{2ab}{k} = \frac{2^{k-2}(ab)^{k-1}}{k!(k-2)!} h_{k,\tau}(a/b) + O_k((a+b)^{2k-3}), \quad (27)$$

$$\frac{\#_\tau(\text{ES}^\pm(a, b)) - \binom{2ab}{k}/k!}{(2ab)^{k-1}} = \frac{h_{k,\tau}(a/b)}{2k!(k-2)!} + o_k(1). \quad (28)$$

Indeed, the $(2, 0)$, $(1, 1)$, and $(0, 2)$ defect classes have respectively aspect factors θ^{-1} , 1, and θ ; their uniform averages are μ_k , $1 - \frac{k}{k-1} \mathbb{E}S$, and μ_k , as computed above. All higher-defect terms have degree at most $2k - 3$. In particular, $h_{k,\tau}(1) = g_k(\tau)$, and exchanging the axes exchanges τ with τ^{-1} and θ with θ^{-1} . The aspect weights separate the two one-axis triple contributions from the mixed contribution. Section 8 uses this separation to optimize the residual after boundary completion and insertions.

For $k = 4$, one direct local evaluation is enough for the comparison needed here:

$$h_{4,1243}(\theta) = -\frac{\theta^{-1} + 1 + \theta}{3}, \quad \frac{|h_{4,1243}(\theta)|}{96} = \frac{\theta^{-1} + 1 + \theta}{288} \geq \frac{1}{96} > \frac{1}{192}.$$

Thus the oriented inserted family of Theorem 8.10 improves every standard fixed-aspect complete core. The square core has the exact formulas below.

7.5 Exact four-point profiles and the published square coefficient

For $k = 4$, direct collection of the composition terms in (14) gives six exact polynomial classes for $m \geq 2$ (some coincide at small m):

$\frac{144}{m^2(m^2-1)} \left(\#_\tau - \frac{1}{24} \binom{2m^2}{4} \right)$	#	τ
$-5(4m^2 - 9)$	1	1234
$-(12m^2 - 13)$	3	1243, 1324, 1432
$4m^2 - 3$	9	1342, 1423, 2143, 2413, 2431, 3142, 3412, 4132, 4231
$-(4m^2 - 5)$	3	2134, 3214, 4321
-11	6	2314, 2341, 3124, 3421, 4123, 4312
$16m^2 - 3$	2	3241, 4213

The table is an exact substitution in the finite rectangle formula, with no asymptotic truncation. It yields

$$\delta_{\text{ES}^\pm(m,m),4} = \begin{cases} \frac{m^2(m^2-1)(16m^2-3)}{144}, & m = 2, 3, \\ \frac{5m^2(m^2-1)(4m^2-9)}{144}, & m \geq 4. \end{cases} \quad (29)$$

In particular the square-grid coefficient is $\Lambda_4 = 5/36$ against m^6 , or $5/288$ against N^3 since $N = 2m^2$. For every $k \geq 2$, (22) gives $g_k(12 \cdots k) = -(k+1)/3$, so $\Lambda_k > 0$.

Remark 7.6 (comparison with the published square coefficient). *In (2), colours $c = +1$ and $c = -1$ correspond respectively to the rotated copies $P(\varepsilon)$ and $P(-\varepsilon)$ of [1, Definitions 5.3 and 5.6]; Section 2 verifies equality of both coordinate orders. Their square side n is our m : both constructions have $N = 2m^2$ and use the centred profile statistic. Our exact formula therefore gives*

$5m^6/36 = 5N^3/288$ as the leading term, rather than the printed $(2/9)m^6$ in [1, Theorem 8]. This bibliographic correction is not used in any bound: the coefficient follows here from Theorem 7.1, Lemma 7.5, and (29). For instance, at $m = 4$ the exact count is $\#_{1234}(\text{ES}^\pm(4, 4)) = 1040$.

8 Bounded insertions and fixed- k realization

The local core vector records what a bounded-prefix row and finitely many insertions can change. For standard sheets, a moment realization attains the resulting LP bound with a nonconstructive family $\Xi_N^{(k, \theta)}$ for fixed k and aspect. A separate periodic orientation gives the explicit $k = 4$ family Σ_N , with 12/17 insertions and constant 1/192. Both optima are relative to their stated completion classes and neither replaces the universal family Π_N .

8.1 The boundary leading vector

For a fixed $0 < \delta < 1$ and integers $0 < r < b$, define the shifted mechanical set

$$R_\delta(b, r) = \left\{ \left\lfloor \frac{(j - \delta)b}{r} \right\rfloor + 1 : 1 \leq j \leq r \right\} \subseteq [b].$$

Its elements are distinct, and for every $0 \leq z \leq b$,

$$\left| |R_\delta(b, r) \cap [z]| - \frac{r}{b}z \right| \leq 2. \quad (30)$$

Indeed, with $y_j = \lfloor (j - \delta)b/r \rfloor + 1$, the condition $y_j \leq z$ is $j < \delta + rz/b$; the number of eligible integers differs from rz/b by at most two. Thus every shifted selector satisfies the hypothesis of Proposition 8.2. The midpoint set $R_{1/2}(b, r)$ is generally distinct from the ceiling selector $R_{b,r}$ used in Section 2.

For $1 \leq t \leq k$ write

$$f_{k,t}^{(b)}(y) = \binom{y-1}{t-1} \binom{b-y}{k-t},$$

and, for $R \subseteq [b]$ with $|R| = r$, put

$$q_{k,t}(b, R) = \frac{1}{b^{k-1}} \left(\sum_{y \in R} f_{k,t}^{(b)}(y) - \frac{r}{b} \binom{b}{k} \right). \quad (31)$$

This is the normalized error from replacing R by density r/b against this rank weight. For example, $R_{1/2}(5, 2) = \{2, 4\}$ gives $q_{4,3} = 1/125$. At scale N^{k-1} it is the only pattern-dependent boundary term; collision terms become uniform after density replacement.

Lemma 8.1 (bounded-prefix Abel estimate). *Let g be supported on an integer interval $I = [u, v]$, and suppose every initial sum on that interval satisfies*

$$\left| \sum_{y=u}^z g(y) \right| \leq D \quad (u \leq z \leq v).$$

Then every weight $W : I \rightarrow \mathbb{R}$ satisfies

$$\left| \sum_{y=u}^v g(y) W(y) \right| \leq D \left(|W(v)| + \sum_{y=u}^{v-1} |W(y+1) - W(y)| \right). \quad (32)$$

If $\sum_I g = 0$, the endpoint term $D|W(v)|$ may be omitted. In particular, if $D = O(1)$, W is a polynomial of degree at most d , and $\|W\|_\infty = O_d(b^d)$ on $I \subseteq [b]$, then the left side is $O_{d,D}(b^d)$, uniformly in the subinterval endpoints. These hypotheses hold for the rank-completion weights below, whose binomial factors have arguments between 0 and b .

Proof. Set $G(z) = \sum_{y=u}^z g(y)$. Summation by parts gives $\sum gW = G(v)W(v) + \sum_{y=u}^{v-1} G(y)(W(y) - W(y+1))$, which proves (32); $G(v) = 0$ removes the endpoint term. A fixed-degree polynomial has $O_d(1)$ monotonicity intervals, so its total variation is $O_d(b^d)$ when its size is $O_d(b^d)$. The same size bound holds for the rank-completion weights because a degree- j binomial factor is at most $b^j/j!$. $\square$

Proposition 8.2 (boundary leading vector). *Fix $k \geq 2$. Suppose $a, b \rightarrow \infty$, $a/b \rightarrow \theta > 0$, and $R \subseteq [b]$, $|R| = r$, satisfies*

$$\sup_{0 \leq z \leq b} \left| |R \cap [z]| - \frac{r}{b}z \right| = O(1).$$

For

$$F = ([a] \times [b]) \cup (\{a+1\} \times R), \quad N = 2|F|,$$

one has, uniformly over $\tau \in \mathfrak{S}_k$,

$$\frac{\#_\tau(\Pi(F)) - \binom{N}{k}/k!}{N^{k-1}} = \frac{h_{k,\tau}(\theta)}{2k!(k-2)!} + \frac{2}{(k-1)!} q_{k,\tau(k)}(b, R) + o(1). \quad (33)$$

Proof. Let C be a fixed bound for the prefix discrepancy. Use the composition-colour-word partition from Theorem 7.1. For $\alpha = (\alpha_1, \dots, \alpha_p)$ put

$$I_\alpha = \{k - \alpha_p + 1, \dots, k\},$$

the last X -block. When a term uses the partial row, this block has abscissa $a+1$. For $\alpha, \beta \models k$, set

$$Q_{\alpha,\beta,\tau} = \{p_\beta(\tau(i)) : i \in I_\alpha\} \subseteq [\ell(\beta)]$$

and, for $Q \subseteq [q]$, set

$$S_Q(b, R) = \sum_{1 \leq y_1 < \dots < y_q \leq b} \prod_{j \in Q} \mathbf{1}_R(y_j).$$

The partial-row contribution of (α, β) is exactly

$$\binom{a}{p-1} w_\tau(\alpha, \beta) S_{Q_{\alpha,\beta,\tau}}(b, R), \quad p = \ell(\alpha), \quad q = \ell(\beta). \quad (34)$$

Indeed, the other $p-1$ abscissae lie in $[a]$, and precisely the ordinate levels meeting the last X -block must lie in R . This condition depends only on the coordinate blocks, so the full colour sum w_τ is retained.

The summand in (34) is $O_k(a^{p-1}b^q)$. Therefore $p+q \leq 2k-2$ contributes $O_k(b^{2k-3}) = o(N^{k-1})$. For the partial-row part with $p = q = k$, $w_\tau(1^k, 1^k) = 2^k$ and $Q = \{\tau(k)\}$, giving

$$2^k \binom{a}{k-1} \sum_{y \in R} f_{k,\tau(k)}^{(b)}(y).$$

The profile average of the sum is $(r/b)\binom{b}{k}$ by Vandermonde; after division by N^{k-1} , the centred remainder is the second term of (33).

It remains to handle $p + q = 2k - 1$, for which $|Q| \leq 2$. Uniformly for fixed $Q \subseteq [q]$ with $|Q| \leq 2$,

$$S_Q(b, R) = \left(\frac{r}{b}\right)^{|Q|} \binom{b}{q} + O_{k,C}(b^{q-1}). \quad (35)$$

For $|Q| = 0$ this is exact. For $|Q| = 1$, apply Lemma 8.1 to $g = \mathbf{1}_R - r/b$, whose prefix sums are bounded by C , and to $W_{q,j}(y) = \binom{y-1}{j-1} \binom{b-y}{q-j}$; its degree is $q - 1$, so the error is $O_{k,C}(b^{q-1})$. If $Q = \{j, \ell\}$ with $j < \ell$, expand each indicator as $r/b + g$. The one- g terms are covered by the same argument. For the two- g term, fix $v = y_\ell$; the inner sum over $u = y_j < v$ has weight

$$U_v(u) = \binom{u-1}{j-1} \binom{v-u-1}{\ell-j-1} \binom{b-v}{q-\ell},$$

with inadmissible binomial coefficients zero. Every interval sum of g has absolute value at most $2C$. The support of U_v is $\{j, \dots, v - (\ell - j)\}$, and on it U_v has degree at most $q - 2$, supremum $O_k(b^{q-2})$, and variation $O_k(b^{q-2})$. Abel summation, retaining its endpoint term, gives uniformly in v

$$\left| \sum_{u < v} g(u) U_v(u) \right| \leq 2C(\|U_v\|_\infty + \text{Var}(U_v)) = O_{k,C}(b^{q-2}).$$

Since $|g(v)| \leq 1$, summing over v gives $O_{k,C}(b^{q-1})$ and proves (35).

The only leading partial-row types are $(p, q) = (k - 1, k)$ and $(k, k - 1)$. In the first, $\beta = 1^k$, α has one part of size two, and $|Q| = \alpha_p \in \{1, 2\}$, independently of τ ; in the second, $\alpha = 1^k$ and $|Q| = 1$. After (35), the first main term is $(r/b)^{\alpha_p}$ times the corresponding complete-row class with the largest abscissa fixed at $a + 1$. For each fixed original x -multiset and set of distinct ordinates, the first part of Lemma 4.1, with its full colour sum, makes this vector pattern-uniform. The second main term is r/b times the analogous class with distinct abscissae and one repeated ordinate, which is uniform by the symmetric part of that lemma. No claim is needed for an individual colour word; (34) retains the complete sum, and the scalar depends only on the composition.

Each replacement error is $O_{k,C}(a^{p-1}b^{q-1}) = O_{k,C}(b^{2k-3})$, and there are finitely many compositions for fixed k . Thus all partial-row main terms other than the clean boundary term are pattern-uniform and all errors are $o(N^{k-1})$. The complete core contributes the first term of (33) by (27). $\square$

Corollary 8.3 (universal-family coefficient). *For every fixed $k \geq 2$, with Λ_k as in Theorem 7.1,*

$$\limsup_{N \rightarrow \infty} \frac{\delta_{\Pi_N, k}}{N^{k-1}} \leq \frac{\Lambda_k}{2^{k-1}} + \frac{3k-1}{k!(k-1)!}.$$

Proof. Lemma 3.3 gives $|q_{k,t}(b, R_{b,r})| \leq 1/(k-1)!$. Proposition 8.2, with $a/b \rightarrow 1$, bounds the even-order limsup by $\Lambda_k/2^{k-1} + 2/((k-1)!)^2$. For odd orders, (12) adds $(k-1)/(k!(k-1)!)$: the term $\delta_{\Pi_{N-1}, k-1}$ is lower order for fixed $k \geq 3$, and $\delta_{\Pi_{N-1}, 1} = 0$. Summing proves the claim. $\square$

Lemma 8.4 (midpoint quadrature). *Suppose $b, r \rightarrow \infty$ and $r/b \rightarrow \lambda \in (0, 1)$ is irrational. For every fixed k and t ,*

$$q_{k,t}(b, R_{1/2}(b, r)) \rightarrow 0.$$

Proof. Let

$$A_z = |R_{1/2}(b, r) \cap [z]| - \frac{r}{b}z \quad (0 \leq z \leq b).$$

Vandermonde and summation by parts give

$$q_{k,t}(b, R_{1/2}(b, r)) = -\frac{1}{b^{k-1}} \sum_{z=1}^{b-1} A_z (f_{k,t}^{(b)}(z+1) - f_{k,t}^{(b)}(z)). \quad (36)$$

Write $\langle x \rangle = x - \lceil x \rceil + 1$. Then $A_z = \frac{1}{2} - \langle \frac{1}{2} + rz/b \rangle$, and the pairs

$$\left(\frac{z}{b}, \left(\frac{1}{2} + \frac{rz}{b} \right) \pmod{1} \right), \quad 1 \leq z < b,$$

are equidistributed in $[0, 1]^2$. Indeed, for fixed integers m, n not both zero the Fourier sum is geometric with ratio $\exp(2\pi i(m + nr)/b)$. For $n \neq 0$ this tends to $\exp(2\pi i n \lambda) \neq 1$, while for $n = 0$ the complete roots-of-unity sum has normalized magnitude $O(1/b)$. The integer endpoints of $\langle \cdot \rangle$ account for $o(b)$ indices, since the reduced denominators tend to infinity.

Put $g_{k,t}(x) = x^{t-1}(1-x)^{k-t}/((t-1)!(k-t)!)$. Uniformly for $0 \leq z < b$,

$$\frac{f_{k,t}^{(b)}(z+1) - f_{k,t}^{(b)}(z)}{b^{k-2}} = g'_{k,t}(z/b) + O_k(1/b).$$

Thus (36) is an equidistributed Riemann sum for $(\frac{1}{2} - u)g'_{k,t}(x)$, whose integral is zero because $\int_0^1 (\frac{1}{2} - u) du = 0$. The uniform remainder and the exceptional indices are $o(1)$, proving the limit. $\square$

Lemma 8.5 (dimension search). *Fix $\lambda \in (0, 1)$. For sufficiently large integers E , put $B = \lfloor \sqrt{E} \rfloor$ and $L = \lceil B^{2/3} \rceil$. Among $b = B, \dots, B + L$, choose the smallest minimizer of $d(E/b, \lambda)$, where $d(u, v) = \min_{m \in \mathbb{Z}} |u - v - m|$, and set $a = \lfloor E/b \rfloor$, $r = E - ab$. Then*

$$a/b \rightarrow 1, \quad r/b \rightarrow \lambda. \quad (37)$$

Proof. Every candidate has $b/B \rightarrow 1$ and $a/b \rightarrow 1$. For $0 \leq j \leq J = \lfloor 2\sqrt{B} \rfloor < L$, put $F_j = E/(B+j) + j$. Uniformly in this range,

$$F_j = E/B + j^2/B + O(B^{-1/2}), \quad F_{j+1} - F_j = O(B^{-1/2}).$$

Since $F_J - F_0 \rightarrow 4$, the first crossing of a level in $\mathbb{Z} + \lambda$ between the endpoints has overshoot $O(B^{-1/2})$. Subtracting the integer j does not change its distance modulo one, so the minimum tends to zero. As $\lambda \in (0, 1)$, this gives the ordinary limit $r/b \rightarrow \lambda$ and eventually $0 < r < b$. $\square$

8.2 Uniform fixed- k insertion expansion

A *point insertion* adjoins an element to both defining total orders without changing the old relative orders. With n old elements, encode its slot by normalized ranks $(u, v) = (x/n, y/n)$, where x, y count old elements preceding it. All slots are allowed, including endpoints and slots inside coordinate blocks or the partial row; total orders resolve displayed ties.

For fixed k , put

$$A_k = 2k!(k-2)!, \quad s_k = ((k-1)!)^2, \quad B_j^{(k)}(u) = \binom{k-1}{j} u^j (1-u)^{k-1-j} \quad (0 \leq j \leq k-1).$$

The following lemma is the uniform insertion statement used by both the explicit four-point construction and the fixed- k theorem.

Lemma 8.6 (uniform fixed- k insertion expansion). *Fix $k \geq 2$. Suppose $a, b \rightarrow \infty$, $a/b \rightarrow \theta > 0$, and*

$$F = ([a] \times [b]) \cup (\{a+1\} \times R), \quad n = 2|F|,$$

where $R \subseteq [b]$ has bounded prefix discrepancy as in Proposition 8.2. Fix M , and make $m \leq M$ point insertions with normalized ranks $(u_j, v_j) \in [0, 1]^2$, or with arbitrary slots having those normalized ranks. If $N = n + m$ is the resulting size, then, uniformly over the list, the slots, and $\tau \in \mathfrak{S}_k$,

$$\begin{aligned} \frac{\#_\tau(\sigma) - \binom{N}{k}/k!}{N^{k-1}} &= \frac{h_{k,\tau}(\theta)}{A_k} + \frac{2}{(k-1)!} q_{k,\tau(k)}(b, R) \\ &+ \sum_{j=1}^m \left[\frac{1}{s_k} \sum_{i=1}^k B_{i-1}^{(k)}(u_j) B_{\tau(i)-1}^{(k)}(v_j) - \frac{1}{k!(k-1)!} \right] + o(1). \end{aligned} \quad (38)$$

For asymptotic purposes, a slot with normalized ranks (u_j, v_j) may, when $m \geq 1$, be represented by the point

$$p_j = \left(\lfloor u_j a \rfloor + \frac{j}{m+1}, \lfloor v_j b \rfloor + \frac{j}{m+1}, 0 \right), \quad 1 \leq j \leq m. \quad (39)$$

The fractional offsets separate the displayed inserted points; arbitrary slots, including endpoints and slots inside coordinate blocks, are handled by the movement estimate in the proof. The old points use the same coordinate keys as in (2).

Proof. For the displayed representatives, configurations containing at least two inserted points number $O_{k,M}(n^{k-2})$; the case $m = 0$ is empty. Fix one inserted point p_j and suppose it has position i in the selected X -order and value position $\tau(i)$ in the selected Y -order. After excluding a partial-row point and all old coordinate collisions, its leading coordinate-set contribution is

$$2^{k-1} \binom{\lfloor u_j a \rfloor}{i-1} \binom{a - \lfloor u_j a \rfloor}{k-i} \binom{\lfloor v_j b \rfloor}{\tau(i)-1} \binom{b - \lfloor v_j b \rfloor}{k-\tau(i)}.$$

The $O(1)$ fractional offsets in (39) are covered by the uniform binomial estimate below. The factor 2^{k-1} counts the independent old colours; once the two coordinate sets and the insertion ranks are fixed, the pattern fixes the cell bijection.

Uniformly for $0 \leq u \leq 1$ and fixed $d \geq 1$,

$$\binom{\lfloor ua \rfloor + O(1)}{d} = \frac{(ua)^d}{d!} + O_d(a^{d-1}),$$

with the analogous estimate for b, v and exact value one when $d = 0$. The absolute estimates include $u, v = 0, 1$. Summing over i and using $N^{k-1} \sim (2ab)^{k-1}$ gives

$$\frac{1}{s_k} \sum_{i=1}^k B_{i-1}^{(k)}(u_j) B_{\tau(i)-1}^{(k)}(v_j) + O_k(a^{-1} + b^{-1}).$$

The omitted repeated-old-coordinate and partial-row configurations total

$$O_k(a^{k-2}b^{k-1} + a^{k-1}b^{k-2} + a^{k-2}rb^{k-2}) = O_{k,\theta}(n^{k-3/2}),$$

uniformly in the cuts, hence are $o(n^{k-1})$ because $r < b$ and $a/b \rightarrow \theta$. The same holds for the multi-insertion configurations. The target changes by

$$\frac{1}{k!} \left(\binom{n+m}{k} - \binom{n}{k} \right) = \frac{m n^{k-1}}{k!(k-1)!} + O_{k,M}(n^{k-2}),$$

which supplies the bracket in (38). Proposition 8.2 supplies the core and row terms, while its Abel estimate gives $q_{k,t}(b, R) = O_k(1)$. This proves the formula for the representatives, uniformly in the macro-coordinates.

For an arbitrary slot, a cut at $x = ua$ has $2b$ old points per complete column and at most $2|R| = O(b)$ partial-row points; its old rank therefore differs from the prescribed rank by $O(a + b)$. The analogous y -cut bound uses column masses $2a$ and at most two extra points. Moving an insertion through $D = O(a+b) = O(\sqrt{n})$ old positions changes at most $\binom{n+m-2}{k-2} = O_{k,M}(n^{k-2})$ configurations per position; all configurations with two inserted points total $O_{k,M}(n^{k-2})$. The movement error is consequently $O_{k,M}(n^{k-3/2}) = o(n^{k-1})$. Finally, perturbing finitely many coincident displayed coordinates by $o(1)$ in normalized units removes all old and new coordinate ties. Continuity of the Bernstein factors preserves the leading term and the endpoint uniformity. $\square$

8.3 Bounded moment realization

We realize any bounded centered correction with a bounded number of insertions. Let

$$\mathbf{1} = (1, \dots, 1)^\top, \quad \mathcal{P}_k = I_k - \frac{1}{k} \mathbf{1} \mathbf{1}^\top, \quad \mathcal{V}_k = \{T \in \mathbb{R}^{k \times k} : T \mathbf{1} = 0, \mathbf{1}^\top T = 0\}.$$

For a finite ordered list $Z = ((u_\ell, v_\ell))_{\ell=1}^m$ in $[0, 1]^2$, put

$$\Phi_k(Z) = \mathcal{P}_k \left(\sum_{\ell=1}^m B^{(k)}(u_\ell) B^{(k)}(v_\ell)^\top \right) \mathcal{P}_k, \quad B^{(k)}(u) = (B_0^{(k)}(u), \dots, B_{k-1}^{(k)}(u))^\top.$$

Then $\Phi_k(Z) \in \mathcal{V}_k$. If $Q = \sum_{\ell=1}^m B^{(k)}(u_\ell) B^{(k)}(v_\ell)^\top$, then $\sum_{i,j} Q_{ij} = m$ and, for every $\tau \in \mathfrak{S}_k$,

$$\sum_{i=1}^k (\mathcal{P}_k Q \mathcal{P}_k)_{i, \tau(i)} = \sum_{i=1}^k Q_{i, \tau(i)} - \frac{m}{k}. \quad (40)$$

Lemma 8.7 (compact centered moment realization). *For every compact set $K \subset \mathcal{V}_k$, there are an integer m and a continuous choice of an m -point list $Z(T)$, $T \in K$, with all coordinates in a fixed compact subset of $(0, 1)^2$, such that*

$$\Phi_k(Z(T)) = T \quad (T \in K).$$

The integer m may be chosen even. There is also an odd integer h and an h -point neutral list Z_{odd} with $\Phi_k(Z_{\text{odd}}) = 0$.

Proof. For general averaging sets see Seymour–Zaslavsky [8]. Take a positive k -node Gauss rule (x_i, w_i) on $[0, 1]$, with distinct interior nodes, exact through degree $2k - 1$. For every sufficiently large L , choose positive integers n_i with $\sum_i n_i = L$ and $n_i/L = w_i + O(1/L)$. Fix one node x_{i_0} . The Jacobian $(dw_i x_i^{d-1})_{1 \leq d \leq k-1, i \neq i_0}$ is an invertible Vandermonde scaling, so the implicit-function theorem adjusts the other nodes by $O(1/L)$ to restore moments $1/(d+1)$ for $1 \leq d \leq k-1$. Degree zero is automatic. The resulting equal-weight list U_L satisfies $\sum_{u \in U_L} B^{(k)}(u) = (L/k) \mathbf{1}$, with support in a fixed interior compact interval. This works for every large L , including odd L .

Fix one such L . The product $Z_0 = U_L \times U_L$ is neutral, with uncentered moment matrix $L^2 \mathbf{1} \mathbf{1}^\top / k^2$. Moving the u -coordinate of one indexed copy at each distinct support pair gives Jacobian columns $B^{(k)'}(x_i) (\mathcal{P}_k B^{(k)}(y_j))^\top$. The vectors $B^{(k)'}(x_i)$ span the zero-sum hyperplane: a vector orthogonal to all of them gives a Bernstein polynomial whose derivative vanishes at k distinct

points, hence is constant. The vectors $\mathcal{P}_k B^{(k)}(y_j)$ span the same hyperplane because the Bernstein evaluation matrix is invertible. Thus the derivative of Φ_k is onto $\mathcal{V}_k$, of dimension $(k-1)^2$. Fixing complementary coordinates, the inverse-function theorem provides a continuous local right inverse near 0 with fixed list size and interior coordinates.

For compact K , choose an even L_0 so large that K/L_0 lies in this neighbourhood. Realize T/L_0 locally and concatenate L_0 copies; by additivity this realizes every $T \in K$ with one fixed even number m of points, continuously in T . Choosing instead a sufficiently large odd L_1 in the first construction gives $Z_{\text{odd}} = U_{L_1} \times U_{L_1}$, with odd size $h = L_1^2$ and $\Phi_k(Z_{\text{odd}}) = 0$. All coordinates remain in one fixed interior compact set. $\square$

8.4 The fixed- k bounded-insertion theorem

Fix $k \geq 4$ and $\theta > 0$. A sequence is a *prescribed-aspect bounded-insertion completion sequence* if deleting at most M inserted points, for one constant M , leaves the two-sheet order $\Pi(F)$ with

$$F = ([a] \times [b]) \cup (\{a+1\} \times R), \quad a, b \rightarrow \infty, \quad a/b \rightarrow \theta,$$

where $R \subseteq [b]$, $|R| = r$, and

$$\sup_{0 \leq z \leq b} \left| |R \cap [z]| - \frac{r}{b} z \right| \leq C$$

for one constant C . Insertion ranks are arbitrary, including endpoints; no geometric separation or limiting location is assumed.

For $D \in \mathbb{R}^{k \times k}$ write $A_D(\tau) = \sum_{i=1}^k D_{i, \tau(i)}$, and define

$$q_k(\theta) = \min_{\substack{D \in \mathbb{R}^{k \times k} \\ \sum_{i,j} D_{ij} = 0}} \max_{\tau \in \mathfrak{S}_k} |h_{k, \tau}(\theta) - A_D(\tau)|.$$

The minimum is attained because these assignment vectors form a finite-dimensional closed subspace of $\mathbb{R}^{k!}$; it is the ℓ_∞ distance from $(h_{k, \tau}(\theta))_\tau$. Replacing D by $\mathcal{P}_k D \mathcal{P}_k$ preserves every assignment sum when $\sum D_{ij} = 0$.

Theorem 8.8 (prescribed-aspect bounded-insertion optimum). *Fix $k \geq 4$ and $\theta > 0$. There is a constant $M_{k, \theta}$ and, for every sufficiently large integer N , a permutation $\Xi_N^{(k, \theta)} \in \mathfrak{S}_N$ obtained from a two-sheet row completion with $a/b \rightarrow \theta$ by at most $M_{k, \theta}$ point insertions such that*

$$\lim_{N \rightarrow \infty} \frac{\delta_{\Xi_N^{(k, \theta)}, k}}{N^{k-1}} = \frac{q_k(\theta)}{2k!(k-2)!}.$$

The family may depend on k and θ ; the construction is nonconstructive. It need not have rational or efficiently computable insertion coordinates. Conversely, every prescribed-aspect bounded-insertion completion sequence has

$$\liminf_{\nu \rightarrow \infty} \frac{\delta_{\pi_\nu, k}}{N_\nu^{k-1}} \geq \frac{q_k(\theta)}{2k!(k-2)!}.$$

This is an optimum only within the displayed completion class, not over all permutations or all aspects.

Proof. Choose a total-sum-zero D attaining $q_k(\theta)$. For the upper construction take the ceiling selector $R = R_{b, r}$ when $r > 0$ and $R = \emptyset$ when $r = 0$, from Section 2. Its prefix discrepancy is at

most one, so Lemma 8.1 and Proposition 8.2 give a constant C'_k such that, for every resulting old order of even size $n = 2|F|$, the matrix E_n with

$$(E_n)_{k,t} = \frac{2}{(k-1)!} q_{k,t}(b, R), \quad (E_n)_{i,t} = 0 \quad (i < k),$$

has uniformly bounded entries. Vandermonde gives $\sum_{t=1}^k q_{k,t}(b, R) = 0$, so E_n has total sum zero and $\mathbf{A}_{E_n}(\tau)$ is exactly the row term in (33).

Let K be a compact ball in $\mathcal{V}_k$ containing all matrices

$$T_n = -s_k \mathcal{P}_k \left(\frac{D}{A_k} + E_n \right) \mathcal{P}_k.$$

Lemma 8.7 supplies one even list size m realizing all $T_n \in K$ and one odd neutral list of size h . The ball is chosen before m, h , so this choice is not circular.

For even N use $n = N - m$; for odd N use $n = N - m - h$. Put $E = n/2$ and choose

$$b = \lfloor \sqrt{E/\theta} \rfloor, \quad a = \left\lfloor \frac{E}{b} \right\rfloor, \quad r = E - ab, \quad R = R_{b,r}$$

with $R = \emptyset$ if $r = 0$. For large sizes, $0 \leq r < b$ and $a/b \rightarrow \theta$, while $|R \cap [z]| = \lfloor rz/b \rfloor$ gives a uniform prefix bound. Realize T_n by m insertions and append the h neutral insertions in the odd case. Perturb the finitely many macro-coordinates by $o(1)$ to make both total orders injective; choose any permutation for the finitely many smaller sizes.

At the old size n , Proposition 8.2 gives, uniformly in τ ,

$$\frac{\#_\tau(\Pi(F)) - \binom{n}{k}/k!}{n^{k-1}} = \frac{h_{k,\tau}(\theta)}{A_k} + \mathbf{A}_{E_n}(\tau) + o(1).$$

By Lemma 8.6, the realization $\Phi_k(Z) = T_n$ changes this vector by $-\mathbf{A}_{D/A_k+E_n}(\tau) + o(1)$; centering does not change the assignment sum of a total-sum-zero matrix. The row term cancels, leaving

$$\frac{h_{k,\tau}(\theta) - \mathbf{A}_D(\tau)}{A_k} + o(1).$$

Since $N = n + O_{k,\theta}(1)$, taking the maximum and then the limit proves the upper equality in the theorem.

For the converse, apply Lemma 8.6 to any sequence in this class. The bounded-prefix hypothesis controls the row term and all errors uniformly, and the bounded insertion list contributes a centered moment matrix of bounded mass. After multiplying it by A_k/s_k and absorbing the row matrix E_n , the leading profile is

$$\frac{h_{k,\tau}(\theta) + \mathbf{A}_{D_\nu}(\tau)}{A_k} + o(1)$$

for a total-sum-zero matrix D_ν (depending on ν and not required to converge). Replacing D_ν by $-D_\nu$ in the definition of $q_k(\theta)$ gives the pointwise lower bound $q_k(\theta)/A_k - o(1)$ on the maximum discrepancy. Taking the liminf proves the converse. $\square$

8.5 Exact LP constants and aspect optimization

For each listed (k, θ) we give matching primal and dual certificates. A matrix D is primal for q if $\sum_{i,j} D_{ij} = 0$ and, for every $\tau \in \mathfrak{S}_k$,

$$\left| h_{k,\tau}(\theta) - \sum_i D_{i,\tau(i)} \right| \leq q.$$

A rational signed vector $(w_\tau)_{\tau \in \mathfrak{S}_k}$ is dual if

$$\sum_{\tau} |w_\tau| \leq 1, \quad \sum_{\tau: \tau(i)=j} w_\tau \text{ is independent of } (i, j), \quad \sum_{\tau} w_\tau h_{k,\tau}(\theta) = q.$$

The constant-marginal condition annihilates assignment sums from every total-sum-zero matrix, so these primal and dual conditions match by ℓ_∞ - ℓ_1 duality. An exact-arithmetic checker enumerates all patterns for each listed (k, θ) over $\mathbb{Q}$ or $\mathbb{Q}(\sqrt{3})$. Its certificate arrays and checker are in the validation repository, file `validation/exact_lp_certificates.json`; from its root, run `python3 -B -m validation.certificate.audit`. The six rows below use the file's (k, θ) keys; the checker verifies these finite certificates, while the all-size realization is the separate content of Theorem 8.8. Put

$$\rho = 2 - \frac{2\sqrt{3}}{3}, \quad q_* = \frac{111 + 5\sqrt{3}}{60}. \quad (41)$$

k	θ	$q_k(\theta)$	$q_k(\theta)/(2k!(k-2)!)$
4	1	$\frac{2}{3}$	$\frac{1}{144}$
5	1	$\frac{5}{4}$	$\frac{1}{1152}$
6	1	2	$\frac{1}{17280}$
7	1	$\frac{62}{21}$	$\frac{31}{12700800}$
8	1	$\frac{109}{28}$	$\frac{109}{1625702400}$
6	ρ	q_*	$q_*/34560$

(42)

For a square-aspect dual, average w_τ with $w_{\tau^{-1}}$; its norm is still at most one and its assignment marginals remain constant. Since inversion swaps T_x, T_y and preserves S , this gives

$$q_k(\theta) \geq q_k(1) + \alpha_k(\theta + \theta^{-1} - 2), \quad \alpha_k = \frac{1}{2} \sum_{\tau} w_\tau (T_x(\tau) + T_y(\tau) - 2\mu_k).$$

The supplied square duals have $\alpha_k = 0, 0, -1/5, 13/42, 3/8$ for $k = 4, \dots, 8$, respectively, so square aspect is optimal in these completion classes for $k = 4, 5, 7, 8$. For $k = 6$, three supplied rational duals give

$$q_6(\theta) \geq \max \left\{ \frac{8}{5} + \frac{1}{3\theta}, \quad \frac{8}{5} + \frac{\theta}{3}, \quad \frac{12}{5} - \frac{\theta + \theta^{-1}}{5} \right\}.$$

Inversion also gives $q_k(\theta) = q_k(1/\theta)$. On $0 < \theta \leq 1$, the first bound decreases and the third increases; they meet only at ρ , where they equal q_* , so their maximum is strictly larger off ρ . Since the exact primal at ρ attains $q_* < 2 = q_6(1)$, the minimizing aspects for $k = 6$ are exactly ρ and $1/\rho$ within the stated completion framework. For the four-pattern design below, write $B = B^{(4)}$ and $K_\tau(u, v) = \frac{1}{36} \sum_{i=1}^4 B_{i-1}(u) B_{\tau(i)-1}(v) - \frac{1}{144}$ for the single-insertion correction.

8.6 Periodic row orientations

The optimum in Theorem 8.8 fixes the standard orientation of both sheets. For four-pattern counts, changing the relative orientation from row to row lowers the attainable constant. Let $\sigma = (\sigma_x)_{x \geq 1}$ be a fixed periodic sequence in $\{-1, 1\}$, with period p and mean $\eta = p^{-1} \sum_{x=1}^p \sigma_x$. Replace the second order in (2) by

$$K_Y^\sigma(x, y, c) = (y, \sigma_x c x, c), \quad K_X(x, y, c) = (x, -c y, c).$$

Write $\Pi^\sigma(F)$ for this order on positive-coordinate cell sets, and put $U(\tau) = |\{i < 4 : |\tau(i+1) - \tau(i)| = 1\}|$. Thus U counts pairs adjacent in both position and value. Define

$$h_\tau^\eta(\theta) = \frac{T_x(\tau) - 4/3}{\theta} + \theta(T_y(\tau) - 4/3) + 1 - \frac{4\eta}{3}S(\tau) - \frac{2(1-\eta)}{3}U(\tau). \quad (43)$$

Lemma 8.9 (oriented leading profile). *For fixed σ and $a/b \rightarrow \theta > 0$, the complete oriented rectangle has centred four-profile $N^3 h_\tau^\eta(\theta)/96 + o(N^3)$, where $N = 2ab$. For its bounded-prefix partial-row completions and bounded insertions, (38) holds at $k = 4$ with $h_{4,\tau}$ replaced by h_τ^η . All other terms and the uniformity in insertion ranks are unchanged.*

Proof. For compositions $\alpha, \beta \models 4$, put $\ell = \ell(\alpha)$, $a_i = p_\alpha(i)$ and $b_i = p_\beta(\tau(i))$. Prescribe signs ε_j for the distinct rows. Let $w_\tau^\varepsilon(\alpha, \beta)$ count colour words with increasing keys $(-c_i b_i, c_i)$ within X -blocks and $(\varepsilon_{a_i} c_i a_i, c_i)$ within Y -blocks, excluding repeated signed points. For each actual row choice $x_1 < \dots < x_\ell \leq a$, set $\varepsilon_j = \sigma_{x_j}$. The exact core profile sums $w_\tau^\varepsilon(\alpha, \beta) \binom{b}{\ell(\beta)}$ over these choices and all compositions. Defect-zero and defect-one weights are pattern-uniform for every prescribed row-sign word: at a single pair the two constraints are interchanged by reversing both colours, and singleton colours are free. Their full contributions therefore vanish after centering, including all lower coefficients of the coordinate-choice factors.

For defect two, average distinct-row signs independently with mean η . The colour sums, aggregated over compositions with the indicated lengths, are

$(\ell(\alpha), \ell(\beta))$	averaged colour sum
$(2, 4)$	$4(T_x + 1)$
$(4, 2)$	$4(T_y + 1)$
$(3, 3)$	$4(9 - \eta S - (1 - \eta)U/2)$.

The one-axis rows follow from the triple and disjoint-pair counts in Lemma 7.5. For the mixed row, use the active endpoints in that proof. Edges with at most one common point have normalized weight 1 if their active endpoints differ, and $1 - \eta h_e(i) v_f(i)$ otherwise. If both endpoints coincide, the two points occupy one cell: the averaged opposite-colour weight is $(1 + \eta)/2$ for a descent and $(1 - \eta)/2$ for an ascent. Summing these nine edge-pair cases gives $9 - \eta S - (1 - \eta)U/2$. The repository's `validation/oriented_four.py` also checks all 24 patterns and the polynomial coefficients in η directly from the tie comparisons.

This independent-sign average gives the leading coefficient for a fixed periodic sequence: for each residue word of ℓ increasing rows, the number of choices is $a^\ell / (\ell! p^\ell) + O_p(a^{\ell-1})$. Multiplying by $\binom{b}{\ell(\beta)}$ makes this error $O_p(N^{5/2})$ in defect two. Larger defects have the same lower order. For a uniform four-pattern, $\mathbb{E}T_x = \mathbb{E}T_y = 4/3$, $\mathbb{E}S = 3/4$, and $\mathbb{E}U = 3/2$, the last identity counting the three adjacent pairs. Center the table, divide by the coordinate factorials and by $(2ab)^3$, and obtain (43)/96.

For the partial row, replace the factor $\binom{a}{p-1}w_\tau$ in (34) by

$$\sum_{x_1 < \dots < x_{p-1} \leq a} w_\tau^{(\sigma_{x_1}, \dots, \sigma_{x_{p-1}}, \sigma_{a+1})}(\alpha, \beta).$$

For $(p, q) = (4, 4)$ the summand is 16 for each row choice; for $(3, 4)$ or $(4, 3)$ it is 8, independently of the row signs and pattern. Thus the clean Bernstein term and the defect-one density replacement in Proposition 8.2 are unchanged. Every remaining term has size $O(a^{p-1}b^q) = O(b^5) = o(N^3)$ because $p + q \leq 6$. For insertions, the leading coordinate-distinct term again ignores signs; collision counts and old-rank displacement bounds depend only on row and column sizes. The proof of Lemma 8.6 therefore gives the asserted uniform expansion, including endpoints and moving ranks. $\square$

8.7 A cubic moment design

Set $u_i = i/3$ for $0 \leq i \leq 3$, and define

$$D = \frac{1}{192} \begin{pmatrix} 51 & 5 & -8 & -48 \\ 18 & -36 & -63 & 81 \\ -63 & -9 & 72 & 0 \\ -6 & 40 & -1 & -33 \end{pmatrix}, \quad \beta_{is} = \frac{3}{4} + \frac{D_{i+1,s+1}}{2} \quad (0 \leq s \leq 3).$$

Each row of D sums to zero. Put

$$\mu_{i1} = \frac{\beta_{i1} + 2\beta_{i2} + 3\beta_{i3}}{3}, \quad \mu_{i2} = \frac{\beta_{i2}}{3} + \beta_{i3}, \quad \mu_{i3} = \beta_{i3},$$

and let V_i consist of the three roots of

$$f_i(z) = z^3 - \mu_{i1}z^2 + \frac{\mu_{i1}^2 - \mu_{i2}}{2}z - \frac{\mu_{i1}^3 - 3\mu_{i1}\mu_{i2} + 2\mu_{i3}}{6}.$$

Each f_i changes sign from $-$ to $+$ on $(1/10, 1/5)$, from $+$ to $-$ on $(2/5, 7/10)$, and from $-$ to $+$ on $(3/4, 19/20)$. These disjoint intervals contain all three roots, which are distinct and lie in $(0, 1)$. Newton identities give

$$\sum_{v \in V_i} v^j = \mu_{ij} \quad (1 \leq j \leq 3), \quad \sum_{v \in V_i} B_s(v) = \beta_{is} \quad (0 \leq s \leq 3).$$

Use each pair (u_i, v) , $v \in V_i$, once, giving 12 insertion locations. List them by increasing i and then increasing root.

For odd orders append the five points in H , in lexicographic order, where

$$\begin{aligned} G_2 &= \left\{ \frac{1}{2} - \frac{1}{2\sqrt{3}}, \frac{1}{2} + \frac{1}{2\sqrt{3}} \right\}, \\ G_3 &= \left\{ \frac{1}{2} - \frac{1}{2\sqrt{2}}, \frac{1}{2}, \frac{1}{2} + \frac{1}{2\sqrt{2}} \right\}, \\ H &= (\{0\} \times G_2) \cup (\{1\} \times G_3). \end{aligned}$$

The identities $\sum_{v \in G_2} B(v) = \frac{1}{2}\mathbf{1}$ and $\sum_{v \in G_3} B(v) = \frac{3}{4}\mathbf{1}$ give $\mathcal{P}Q_H\mathcal{P} = 0$. The mass-5 identity then gives $\sum_{(u,v) \in H} K_\tau(u, v) = \frac{1}{36}(\frac{1}{2} + \frac{3}{4}) - \frac{5}{144} = 0$, so this padding is neutral for every pattern.

Here is the exact moment calculation. With $A_{s+1,i+1} = B_s(i/3)$ for $0 \leq s, i \leq 3$,

$$27A = \begin{pmatrix} 27 & 8 & 1 & 0 \\ 0 & 12 & 6 & 0 \\ 0 & 6 & 12 & 0 \\ 0 & 1 & 8 & 27 \end{pmatrix}, \quad AD = \frac{1}{32}C, \quad C = \begin{pmatrix} 9 & -1 & -4 & -4 \\ -1 & -3 & -2 & 6 \\ -4 & -2 & 3 & 3 \\ -4 & 6 & 3 & -5 \end{pmatrix}.$$

The last identity follows by multiplying the displayed integer matrices: $(27A)(192D) = 162C$. Write $\mathcal{P} = I_4 - \mathbf{1}\mathbf{1}^\top/4$, where $\mathbf{1} = (1, 1, 1, 1)^\top$. The 12-point moment matrix is $Q = A(\frac{3}{4}\mathbf{1}\mathbf{1}^\top + D/2)$, so $\mathcal{P}Q\mathcal{P} = C/64$. Adding the neutral rule preserves this identity. By (40), both lists have the total correction

$$\sum_j K_\tau(u_j, v_j) = \frac{1}{2304} \sum_{i=1}^4 C_{i,\tau(i)}. \quad (44)$$

Choose the period-eight orientation $(-, +, -, +, +, -, +, +)$, with mean $\eta = 1/4$. Use $m = 12$ when N is even and $m = 17$ when N is odd. Set $\Sigma_N = \Pi_N$ if $N - m < 100$. Otherwise apply Lemma 8.5 to $E = (N - m)/2$ with $\lambda = 3 - \sqrt{5}$. Only its irrationality is needed for the profile limit. If $r = 0$, again use Π_N ; this occurs only finitely often. In all other cases use the oriented order $\Pi^\sigma(F)$ with row $R_{1/2}(b, r)$ and insert the listed m points by (39), keeping one global index j for all locations, including the five appended points at odd orders. The final point count is exactly N .

Theorem 8.10 (all-size four-point constant). *The explicit family (Σ_N) satisfies*

$$\lim_{N \rightarrow \infty} \frac{\delta_{\Sigma_N, 4}}{N^3} = \frac{1}{192}, \quad \limsup_{N \rightarrow \infty} \frac{\delta_4(N)}{N^3} \leq \frac{1}{192}.$$

Conversely, let $\pi_\nu \in \mathfrak{S}_{N_\nu}$, $N_\nu \rightarrow \infty$, be row completions with a fixed periodic orientation, a fixed limiting aspect $\theta > 0$, bounded prefix discrepancy and at most M insertions, where M is fixed. Then

$$\liminf_{\nu \rightarrow \infty} \frac{\delta_{\pi_\nu, 4}}{N_\nu^3} \geq \frac{1}{192}.$$

This optimum allows all fixed periods and positive limiting aspects, but is not asserted over arbitrary permutations.

Proof. The dimension search gives $a/b \rightarrow 1$ and $r/b \rightarrow 3 - \sqrt{5}$. By Lemmas 8.4 and 8.9, the row term vanishes; (44) gives, for both parities,

$$L_\tau = \frac{h_\tau^{1/4}(1)}{96} + \frac{1}{2304} \sum_{i=1}^4 C_{i,\tau(i)}.$$

Substitution in the local formula gives the following complete ledger:

$2304L_\tau$	τ
-12	1243, 1324, 2143, 3412, 4231
-8	4321
-4	2341, 3421, 4123, 4312
0	1234, 1432, 2314, 2431, 3124, 3214, 4132
12	1342, 1423, 2134, 2413, 3142, 3241, 4213

The limiting maximum is $12/2304 = 1/192$.

For optimality, define the ten-pattern functional

$$16\mathcal{W}(z) = -2z_{1243} + 2z_{1342} + 2z_{1423} + 2z_{2134} - z_{2143} \\ + z_{2413} + z_{3142} + 2z_{3241} - z_{3412} + 2z_{4213}.$$

Its coefficient ℓ_1 norm is 1, and every position–value pair has coefficient sum $1/8$. Hence it annihilates assignment sums of total-sum-zero matrices, including the centred row and insertion corrections. Directly,

$$\mathcal{W}(1) = \frac{1}{2}, \quad \mathcal{W}(T_x) = \mathcal{W}(T_y) = \frac{3}{4}, \quad \mathcal{W}(S) = \frac{1}{8}, \quad \mathcal{W}(U) = \frac{1}{4}.$$

Thus (43) gives

$$\mathcal{W}((h_\tau^\eta(\theta))_\tau) = \frac{1}{3} + \frac{\theta + \theta^{-1}}{12} \geq \frac{1}{2},$$

independently of η . Lemma 8.9 and $\|\mathcal{W}\|_1 = 1$ give the lower bound $1/(2 \cdot 96) = 1/192$. The ledger attains this bound. $\square$

Concluding remarks. The exponent-optimal family is k -independent, while completion constants depend on the allowed orientations. The relative-balancing range of this family is sharp. Global constants, optimal aspects beyond the tabulated orders, and growing-pattern ranges attainable by other families remain open.

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Data availability. No external datasets are used. The validation source is at <https://github.com/AutoclickerI/near-balanced-permutations/tree/v1.1.0>. It includes exact primal/dual certificates over $\mathbb{Q}$ and $\mathbb{Q}(\sqrt{3})$ and finite regression checkers; the analytic proofs are in the article. From the repository root run `bash validation/run.sh`.

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